

Appendix: GROMACS Preparation Table

This table contains the 173 retained molecular-dynamics systems listed in `accepted173.txt`, in retained-membership order.

Condition interpretation. We transcribed pH, temperature, D₂O fraction and buffer or salt composition from BMRB and, when needed, linked PDB records. Of the retained records, 167 were labelled pH and six pH*; none was labelled pD. For pH* readings only, we used $pH_{\text{eff}} = pH_{\text{recorded}} + 0.40f_{D_2O}$. Scaling the pure-D₂O correction by the recorded fraction was a project interpolation. Values labelled pH were used as deposited, including four near-pure or pure-D₂O records. D₂O affected condition interpretation only; the simulated solvent remained ordinary TIP3P water.

State and ion assignment. PROPKA3 supplied structure-based pK_a estimates, and Henderson–Hasselbalch fractions were used to choose one fixed protonation state per system. Neutral-His choice was delegated to `pdb2gmx`; cysteine/disulfide states and termini used explicit project rules, while Arg and Tyr retained their standard states. Ionic strength was calculated as $I = \frac{1}{2} \sum_i c_i z_i^2$. Directly parameterised ions were retained where possible, whereas unsupported solution contributions were represented by an NaCl ionic-strength proxy. After solvation, target concentrations were converted to neutral integer ion counts, so the realised box concentration approximated the bulk target. Prepared residue names, disulfides, termini and OXT were applied and the resulting structure and charge were checked before topology generation.

Table 1–GROMACS: Experimental conditions, preparation decisions, calculated solution properties, and final GROMACS systems for the 173 retained molecular-dynamics proteins.

Record	Provided conditions	Protein preparation	R calculations		GROMACS
BMRB 68; PDB 1UBI, 1UBQ	BMRB: ubiquitin; pH 4.7; 323 K; no D ₂ O fraction listed; no solution components listed.	GLU18→GLH; GLU24→GLH; HIS68→HIP; <i>Termini</i> : NMET1/CGLY76; <i>OXT</i> : present.	<i>R</i>	$pH_{\text{eff}} = 4.70$	Solute charge +3; ions 3 Cl [−] ; water 18 791; net charge 0.
			<i>HH</i>	25 titratable; 3 non-default; 4 marginal	
			<i>Ion plan</i>	neutralisation only (genion -conc 0); no salt target from listed components	
BMRB 4082	BMRB: profilin [U- ¹⁵ N], [U- ¹³ C]; 2.0 mM; pH 6.4; 293 K; 90% H ₂ O/10% D ₂ O; 10 mM phosphate, 25 mM KCl, 20 mM β-mercaptoethanol, 1.0 mM EDTA.	HIS119→HIE; HIS133→HIE; <i>Retained</i> : CYS16, CYS70, and CYS127 free thiols; <i>Termini</i> : NALA1/CTYR139; <i>OXT</i> : present.	<i>R</i>	$pH_{\text{eff}} = 6.40$	Solute charge +2; ions 10 K ⁺ , 7 Na ⁺ , 19 Cl [−] ; water 20 746; net charge 0.
			<i>HH</i>	39 titratable; 0 non-default; 1 marginal	
			<i>Target</i>	$I = 42.63$ mM, from KCl 25.00, phosphate 12.68, and EDTA 4.95	
			<i>Proxy</i>	phosphate and EDTA as 17.63 mM NaCl	
			<i>Achieved</i>	$I_{\text{bulk}} = 43.09$ mM (genion : 7 NA; 10 K, 19 CL)	
BMRB 4095; PDB 1AUU	BMRB: sacB mRNA-binding domain [U- ¹⁵ N]; 3 mM; pH 5.2; 301 K; no D ₂ O fraction listed; $I = 300$ mM; no solution components listed. PDB: pH 6; 297 K; I not listed.	HIS11→HIP; <i>Termini</i> : NGLY1/CTYR57; <i>OXT</i> : present.	<i>Final I</i>	$I = 45.63$ mM after neutralisation	Solute charge +5; ions 123 Na ⁺ , 128 Cl [−] ; water 21 533; net charge 0.
			<i>R</i>	$pH_{\text{eff}} = 5.20$	
			<i>HH</i>	23 titratable; 1 non-default; 0 marginal	
			<i>Target</i>	$I = 300.00$ mM, from a deposited scalar value	
			<i>Proxy</i>	deposited ionic-strength scalar as 300.00 mM NaCl	
BMRB 4121; PDB 1BW5	BMRB: Isl-1 homeodomain; 0.8 mM; pH 6.0; 281 K; 90% H ₂ O/10% D ₂ O; 25 mM NaCl, 25 mM Na ₂ HPO ₄ /NaH ₂ PO ₄ , 10 mM DTT. PDB: pH 6.0; 281 K; $I = 50$ mM Na ⁺ .	HIS16→HIP; <i>Retained</i> : CYS21 and CYS56 free thiols; <i>Termini</i> : NMET1/CLYS66; <i>OXT</i> : present.	<i>Achieved</i>	$I_{\text{bulk}} = 301.05$ mM (genion : 123 NA, 128 CL)	Solute charge +12; ions 19 Na ⁺ , 31 Cl [−] ; water 18 608; net charge 0.
			<i>Final I</i>	$I = 307.17$ mM after neutralisation	
			<i>R</i>	$pH_{\text{eff}} = 6.00$	
			<i>HH</i>	24 titratable; 1 non-default; 1 marginal	
			<i>Target</i>	$I = 52.90$ mM, from NaCl 25.00 and phosphate 27.90	
BMRB 4237; PDB 1EV0	BMRB: MinE [U- ¹⁵ N]; 2.9–3.5 mM; pH 5.7; 298 K; no D ₂ O fraction listed; $I = 75$ mM. PDB: 20 mM sodium phosphate, 50 mM NaCl, 15 mM DTT, 1 mM EDTA, 0.02% NaN ₃ , 1 mM AEBSF.	HIS7→HIP; <i>Retained</i> : CYS21 free thiol; <i>Termini</i> : NARG1/CLYS58; <i>OXT</i> : present.	<i>Proxy</i>	phosphate as 27.90 mM NaCl	Solute charge −6; ions 43 Na ⁺ , 37 Cl [−] ; water 26 165; net charge 0.
			<i>Achieved</i>	$I_{\text{bulk}} = 54.16$ mM (genion : 19 NA, 31 CL)	
			<i>Final I</i>	$I = 71.27$ mM after neutralisation	
			<i>R</i>	$pH_{\text{eff}} = 5.70$	
			<i>HH</i>	24 titratable; 1 non-default; 0 marginal	
BMRB 4254; PDB 1L3G	BMRB: MLU1-BOX binding protein [U- ¹⁵ N]; 0.8–1.3 mM; pH 7.6; 288 K; no D ₂ O fraction listed. PDB: 40 mM sodium phosphate, 100 mM NaCl.	HIS19→HIP; HIS37→HIE; HIS62→HID; HIS116→HIE; HIS117→HIE; HIS118→HIE; HIS131→HIE; HIS132→HIE; HIS133→HIE; HIS134→HIE; <i>Termini</i> : NSER1/CHIE135; <i>OXT</i> : present.	<i>Target</i>	$I = 75.00$ mM, from phosphate 21.19, NaCl 50.00, and EDTA 3.81	Solute charge +7; ions 134 Na ⁺ , 141 Cl [−] ; water 36 207; net charge 0.
			<i>Proxy</i>	phosphate and EDTA as 25.00 mM NaCl	
			<i>Achieved</i>	$I_{\text{bulk}} = 74.92$ mM (genion : 43 NA, 37 CL)	
			<i>Final I</i>	$I = 80.99$ mM after neutralisation	
			<i>R</i>	$pH_{\text{eff}} = 7.60$	
			<i>HH</i>	51 titratable; 0 non-default; 1 marginal	
			<i>Target</i>	$I = 196.84$ mM, from phosphate 96.84 and NaCl 100.00	
			<i>Proxy</i>	phosphate as 96.84 mM NaCl	
			<i>Achieved</i>	$I_{\text{bulk}} = 197.30$ mM (genion : 134 NA, 141 CL)	
			<i>Final I</i>	$I = 202.45$ mM after neutralisation	

Record	Provided conditions	Protein preparation	R calculations		GROMACS
BMRB 4271; PDB 1JFJ	BMRB: Entamoeba histolytica CaBP; 3–4 mM; [U- ¹³ C], [U- ¹⁵ N]; pH 6.0; 308 K; no D ₂ O fraction listed; <i>I</i> = 50 mM. PDB: 30 mM deuterated Tris buffer, 30 mM CaCl ₂ .	HIS104→HID; <i>Termini</i> : NMET1/CLEU134; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i>	$pH_{\text{eff}} = 6.00$ 50 titratable; 0 non-default; 1 marginal <i>I</i> = 119.74 mM, from deuterated Tris 29.74 and CaCl ₂ 90.00 deuterated Tris as 29.74 mM NaCl <i>I</i> _{bulk} = 116.77 mM (genion : 14 NA; 18 CA, 40 CL) <i>Final I</i> <i>I</i> = 140.12 mM after neutralisation	Solute charge −10; ions 18 Ca ²⁺ , 14 Na ⁺ , 40 Cl [−] ; water 23 529; net charge 0.
BMRB 4296; PDB 1MJC, 3MEF	BMRB: Cold shock protein A [U- ¹³ C], [U- ¹⁵ N]; 1.0–3.0 mM; pH* 6.0; 303 K; 95% H ₂ O/5% D ₂ O; 50 mM KH ₂ PO ₄ , 0.1 mM NaEDTA, 1 mM NaN ₃ .	HIS33→HID; <i>Termini</i> : NMET1/CLEU70; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i>	$pH_{\text{eff}} = 6.02$ 18 titratable; 0 non-default; 1 marginal <i>I</i> = 50.43 mM, from KH ₂ PO ₄ 50.00 and EDTA 0.43 KH ₂ PO ₄ and EDTA as 50.43 mM NaCl <i>I</i> _{bulk} = 51.86 mM (genion : 18 NA, 17 CL) <i>Final I</i> <i>I</i> = 53.38 mM after neutralisation	Solute charge −1; ions 18 Na ⁺ , 17 Cl [−] ; water 17 417; net charge 0.
BMRB 4301; PDB 1CFE	BMRB: Pathogenesis-related protein P14a; 1.5 mM; pH 5.5; 303 K; 90% H ₂ O/10% D ₂ O; no salts, buffers, chelators, or reductants listed.	HIS11→HIP; HIS48→HIP; HIS93→HIP; <i>Bonded</i> : CYS44–CYS112, CYS85–CYS91, and CYS107–CYS121 S–S; residues CYX; <i>Termini</i> : NGLN1/CTYR135; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Ion plan</i>	$pH_{\text{eff}} = 5.50$ 39 titratable; 8 non-default; 0 marginal neutralisation only (genion -conc 0); no salt target from listed components	Solute charge +9; ions 9 Cl [−] ; water 20 872; net charge 0.
BMRB 4348; PDB 1NTC	BMRB: Nitrogen regulatory protein C [U- ¹³ C], [U- ¹⁵ N]; 0.5–1.5 mM; pH 6.1; 308 K; 20 mM sodium phosphate, 50 mM KCl, 0.1 mM EDTA, 0.02% w/v NaN ₃ . PDB: pH 6.1; 307 K; 90% H ₂ O/10% D ₂ O; <i>I</i> = 70 mM.	HIS18→HIP; HIS40→HIP; HIS62→HIP; HIS66→HIP; <i>Termini</i> : NMET1/CGLU91; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i>	$pH_{\text{eff}} = 6.10$ 25 titratable; 4 non-default; 2 marginal <i>I</i> = 73.32 mM, from phosphate 22.88, KCl 50.00, and EDTA 0.44 phosphate and EDTA as 23.32 mM NaCl <i>I</i> _{bulk} = 73.14 mM (genion : 23 NA; 49 K, 72 CL) <i>Final I</i> <i>I</i> = 73.14 mM after neutralisation	Solute charge 0; ions 49 K ⁺ , 23 Na ⁺ , 72 Cl [−] ; water 52 871; net charge 0.
BMRB 4421	BMRB: PhoB DNA-binding domain [U- ¹³ C], [U- ¹⁵ N]; 1.0–2.0 mM; pH 6.8; 300 K; no D ₂ O fraction listed; no solution components listed.	HIS19→HIE; HIS39→HIE; HIS44→HID; HIS57→HIE; HIS73→HID; HIS86→HIE; <i>Termini</i> : NMET1/CPHE104; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Ion plan</i>	$pH_{\text{eff}} = 6.80$ 38 titratable; 0 non-default; 0 marginal neutralisation only (genion -conc 0); no salt target from listed components	Solute charge −2; ions 2 Na ⁺ ; water 21 822; net charge 0.
BMRB 4430	BMRB: calyculin (rabbit) [U- ¹³ C], [U- ¹⁵ N]; 1.0–2.0 mM; pH 7.0; 300 K; no D ₂ O fraction listed; 50 mM Tris-d11, 0.05% NaN ₃ .	HIS17→HIE; HIS27→HIE; <i>Termini</i> : NMET1/CGLY90; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i>	$pH_{\text{eff}} = 7.00$ 33 titratable; 0 non-default; 0 marginal <i>I</i> = 53.68 mM, from Tris-d11 45.99 and NaN ₃ 7.69 Tris-d11 and NaN ₃ as 53.68 mM NaCl <i>I</i> _{bulk} = 53.90 mM (genion : 24 NA, 21 CL) <i>Final I</i> <i>I</i> = 57.75 mM after neutralisation	Solute charge −3; ions 24 Na ⁺ , 21 Cl [−] ; water 20 726; net charge 0.
BMRB 4493; PDB 1UD7	BMRB/PDB: ubiquitin core mutant 1D7 [U- ¹³ C], [U- ¹⁵ N]; 2 mM; pH 5.8; 303 K; no D ₂ O fraction listed; 25 mM sodium acetate-D3, 0.02% NaN ₃ .	HIS68→HIP; <i>Termini</i> : NMET1/CGLY76; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i>	$pH_{\text{eff}} = 5.80$ 25 titratable; 1 non-default; 1 marginal <i>I</i> = 25.00 mM, from sodium acetate-D3 25.00 sodium acetate-D3 as 25.00 mM NaCl <i>I</i> _{bulk} = 25.98 mM (genion : 8 NA, 9 CL) <i>Final I</i> <i>I</i> = 27.60 mM after neutralisation	Solute charge +1; ions 8 Na ⁺ , 9 Cl [−] ; water 16 566; net charge 0.
BMRB 4510; PDB 1CXW	BMRB: second fibronectin type II module from human matrix metalloproteinase 2; 1.3 mM; pH 5.2; 298 K; 90% H ₂ O/10% D ₂ O; <i>I</i> = 0 M; no salts, buffers, chelators, or reductants listed.	<i>Bonded</i> : CYS15–CYS41 and CYS29–CYS56 S–S; residues CYX; <i>Termini</i> : NTHR1/CALA60; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Ion plan</i>	$pH_{\text{eff}} = 5.20$ 25 titratable; 4 non-default; 0 marginal neutralisation only (genion -conc 0); no salt target from listed components	Solute charge −2; ions 2 Na ⁺ ; water 13 303; net charge 0.
BMRB 4560	BMRB: T4MOD [U- ¹³ C], [U- ¹⁵ N]; 1.1 mM; pH 7.0; 298 K; no D ₂ O fraction listed; no solution components listed.	HIS9→HIE; <i>Termini</i> : NSER1/CMET102; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Ion plan</i>	$pH_{\text{eff}} = 7.00$ 33 titratable; 0 non-default; 0 marginal neutralisation only (genion -conc 0); no salt target from listed components	Solute charge −9; ions 9 Na ⁺ ; water 20 075; net charge 0.
BMRB 4581; PDB 1D1O	BMRB/PDB: Calbindin D9k N56A; 4 mM; pH 6.0; 300 K; no D ₂ O fraction listed; <i>I</i> = 0 mM; no solution components listed.	<i>Termini</i> : NLYS1/CGLN75; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Ion plan</i>	$pH_{\text{eff}} = 6.00$ 29 titratable; 0 non-default; 0 marginal neutralisation only (genion -conc 0); no salt target from listed components	Solute charge −7; ions 7 Na ⁺ ; water 13 237; net charge 0.

Record	Provided conditions	Protein preparation	R calculations		GROMACS
BMRB 4590; PDB 1EIG	BMRB/PDB: Eotaxin-2 [U- ¹³ C], [U- ¹⁵ N]; 1 mM; pH 4.1; 298 K; 90% H ₂ O/10% D ₂ O; 20 mM NaAc/acetate; <i>I</i> = 20 mM.	<i>Bonded</i> : CYS7–CYS32 and CYS8–CYS48 S–S; <i>R</i> residues CYX; <i>Termini</i> : NVAL1/CARG73; <i>OXT</i> : present.	<i>R</i>	$pH_{\text{eff}} = 4.10$	Solute charge +10; ions 2 Na ⁺ , 12 Cl [−] ; water 23 084; net charge 0.
			<i>HH</i>	25 titratable; 4 non-default; 1 marginal	
			<i>Target</i>	<i>I</i> = 3.59 mM, from acetate	
			<i>Proxy</i>	NaAc as 3.59 mM NaCl	
			<i>Achieved</i>	<i>I</i> _{bulk} = 4.63 mM (genion : 2 NA, 12 CL)	
			<i>Final I</i>	<i>I</i> = 16.21 mM after neutralisation	Solute charge −9; ions 24 Na ⁺ , 15 Cl [−] ; water 15 055; net charge 0.
			<i>R</i>	$pH_{\text{eff}} = 5.80$	
			<i>HH</i>	24 titratable; 0 non-default; 0 marginal	
			<i>Target</i>	<i>I</i> = 53.73 mM, from phosphate	
			<i>Proxy</i>	phosphate as 53.73 mM NaCl	
			<i>Achieved</i>	<i>I</i> _{bulk} = 53.36 mM (genion : 24 NA, 15 CL)	Solute charge +1; ions 18 Na ⁺ , 19 Cl [−] ; water 18 768; net charge 0.
			<i>Final I</i>	<i>I</i> = 69.37 mM after neutralisation	
			<i>R</i>	$pH_{\text{eff}} = 5.80$	
			<i>HH</i>	25 titratable; 1 non-default; 1 marginal	
			<i>Target</i>	<i>I</i> = 51.87 mM, from sodium acetate-D3	
				25.00 and phosphate 26.87	
			<i>Proxy</i>	sodium acetate-D3 and phosphate as 51.87 mM NaCl	
			<i>Achieved</i>	<i>I</i> _{bulk} = 50.61 mM (genion : 18 NA, 19 CL)	
			<i>Final I</i>	<i>I</i> = 52.02 mM after neutralisation	
			<i>R</i>	$pH_{\text{eff}} = 6.40$	
			<i>HH</i>	65 titratable; 3 non-default; 1 marginal	Solute charge −1; ions 24 K ⁺ , 12 Na ⁺ , 35 Cl [−] ; water 24 214; net charge 0.
			<i>Target</i>	<i>I</i> = 75.36 mM, from KCl 50.00 and phosphate 25.36	
			<i>Proxy</i>	phosphate as 25.36 mM NaCl	
			<i>Achieved</i>	<i>I</i> _{bulk} = 75.15 mM (genion : 12 NA; 24 K, 35 CL)	
			<i>Final I</i>	<i>I</i> = 76.22 mM after neutralisation	
			<i>R</i>	$pH_{\text{eff}} = 6.50$	Solute charge +3; ions 108 Na ⁺ , 111 Cl [−] ; water 31 377; net charge 0.
			<i>HH</i>	32 titratable; 0 non-default; 2 marginal	
			<i>Target</i>	<i>I</i> = 183.16 mM, from NaCl 150.00 and phosphate 33.16	
			<i>Proxy</i>	phosphate as 33.16 mM NaCl	
			<i>Achieved</i>	<i>I</i> _{bulk} = 183.08 mM (genion : 108 NA, 111 CL)	
			<i>Final I</i>	<i>I</i> = 185.62 mM after neutralisation	Solute charge +5; ions 215 Na ⁺ , 220 Cl [−] ; water 21 472; net charge 0.
			<i>R</i>	$pH_{\text{eff}} = 6.00$	
			<i>HH</i>	29 titratable; 2 non-default; 1 marginal	
			<i>Target</i>	<i>I</i> = 522.32 mM, from NaCl 500.00 and phosphate 22.32	
			<i>Proxy</i>	phosphate as 22.32 mM NaCl	
			<i>Achieved</i>	<i>I</i> _{bulk} = 521.41 mM (genion : 215 NA, 220 CL)	
			<i>Final I</i>	<i>I</i> = 527.48 mM after neutralisation	
			<i>R</i>	$pH_{\text{eff}} = 6.50$	
			<i>HH</i>	43 titratable; 0 non-default; 2 marginal	
			<i>Target</i>	<i>I</i> = 100.00 mM, from KCl 100.00	
			<i>Achieved</i>	genion -conc 0.100	Solute charge −3; ions 62 K ⁺ , 34 Na ⁺ , 93 Cl [−] ; water 31 112; net charge 0.
			<i>R</i>	$pH_{\text{eff}} = 6.20$	
			<i>HH</i>	40 titratable; 1 non-default; 1 marginal	
			<i>Target</i>	<i>I</i> = 158.90 mM, from KCl 100.00 and phosphate 58.90	
			<i>Proxy</i>	phosphate as 58.90 mM NaCl	
			<i>Achieved</i>	<i>I</i> _{bulk} = 158.68 mM (genion : 34 NA; 62 K, 93 CL)	
			<i>Final I</i>	<i>I</i> = 162.11 mM after neutralisation	

Record	Provided conditions	Protein preparation	R calculations		GROMACS
BMRB 4869; PDB 1LWM	BMRB: non-histone protein 6A [U- ¹³ C], [U- ¹⁵ N]; 1.0 mM; pH 5.5; 293 K; no D ₂ O fraction listed; <i>I</i> = 100 mM. PDB: 10 mM sodium phosphate, 100 mM NaCl.	<i>Termini</i> : NMET1/CALA93; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i>	$pH_{\text{eff}} = 5.50$ 41 titratable; 0 non-default; 0 marginal <i>I</i> = 110.38 mM, from NaCl 100.00 and sodium phosphate 10.38 sodium phosphate as 10.38 mM NaCl <i>I</i> _{bulk} = 109.98 mM (genion : 98 NA, 106 CL) <i>Final I</i> <i>I</i> = 114.47 mM after neutralisation	Solute charge +8; ions 98 Na ⁺ , 106 Cl ⁻ ; water 46 889; net charge 0.
BMRB 4935; PDB 1KVN, 1KVV	BMRB: Archaeoglobus fulgidus SRP19 [U-99% ¹³ C], [U-98% ¹⁵ N]; 1.8–2.2 mM; pH 6.0; 300 K; 99.88% D ₂ O; 50 mM NaCl, 25 mM KH ₂ PO ₄ ; <i>I</i> = 75 mM.	HIS35→HIP; <i>Termini</i> : NMET1/CLYS104; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i>	$pH_{\text{eff}} = 6.00$ 52 titratable; 1 non-default; 1 marginal <i>I</i> = 75.00 mM, from NaCl 50.00 and KH ₂ PO ₄ 25.00 KH ₂ PO ₄ as 25.00 mM NaCl <i>I</i> _{bulk} = 75.09 mM (genion : 45 NA, 63 CL) <i>Final I</i> <i>I</i> = 90.10 mM after neutralisation	Solute charge +18; ions 45 Na ⁺ , 63 Cl ⁻ ; water 31 960; net charge 0.
BMRB 4976; PDB 1I2V	BMRB/PDB: heliomicin-LL; 4 mM; pH 4.3; 293 K; no D ₂ O fraction listed; 40 mM sodium acetate buffer; <i>I</i> = 40 mM.	GLU21→GLH; GLU43→GLH; HIS31→HIP; <i>Bonded</i> : CYS7–CYS32, CYS18–CYS40, and CYS22–CYS42 S–S; residues CYX; <i>Termini</i> : NASP1/CTHR44; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 4.30$ 16 titratable; 9 non-default; 2 marginal <i>I</i> = 10.30 mM, from sodium acetate 10.30 sodium acetate as 10.30 mM NaCl <i>I</i> _{bulk} = 11.23 mM (genion : 3 NA, 5 CL) <i>I</i> = 14.97 mM after neutralisation	Solute charge +2; ions 3 Na ⁺ , 5 Cl ⁻ ; water 13 733; net charge 0.
BMRB 4981; PDB 1G4F	BMRB: beta2-glycoprotein I domain V [U- ¹⁵ N]; 1.0 mM; pH 6.0; 298 K; <i>I</i> = 0.12 M. PDB: 20 mM sodium phosphate buffer, 90% H ₂ O/10% D ₂ O.	HIS33→HIP; HIS70→HIE; <i>Bonded</i> : CYS5–CYS56, CYS41–CYS66, and CYS48–CYS86 S–S; residues CYX; <i>Termini</i> : NTHR1/CCYX86; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i>	$pH_{\text{eff}} = 6.00$ 39 titratable; 7 non-default; 2 marginal <i>I</i> = 120.00 mM, from a deposited scalar value	Solute charge +7; ions 56 Na ⁺ , 63 Cl ⁻ ; water 24 518; net charge 0.
BMRB 5027; PDB 1J8K	BMRB: human fibronectin EDA [U- ¹⁵ N], [U- ¹³ C]; 2.0 mM; pH 7.0; 293 K; 20 mM sodium phosphate, 400 mM Na ₂ SO ₄ . PDB: 90% H ₂ O/10% D ₂ O; <i>I</i> = 400 mM as sodium sulfate.	HIS44→HIE; HIS76→HIE; <i>Termini</i> : NASN1/CALA94; <i>OXT</i> : present.	<i>Proxy</i> <i>Achieved</i> <i>Final I</i> <i>R</i> <i>HH</i> <i>Target</i>	sodium phosphate as 120.00 mM NaCl <i>I</i> _{bulk} = 120.39 mM (genion : 56 NA, 63 CL) <i>I</i> = 127.91 mM after neutralisation $pH_{\text{eff}} = 7.00$ 29 titratable; 0 non-default; 0 marginal <i>I</i> = 1235.26 mM, from Na ₂ SO ₄ 1200.00 and phosphate 35.26 Na ₂ SO ₄ and sodium phosphate as 1235.26 mM NaCl <i>I</i> _{bulk} = 1235.70 mM (genion : 568 NA, 557 CL) <i>Final I</i> <i>I</i> = 1247.90 mM after neutralisation	Solute charge −11; ions 568 Na ⁺ , 557 Cl ⁻ ; water 22 741; net charge 0.
BMRB 5031; PDB 1M2F	BMRB: N-terminal KaiA domain [U- ¹³ C], [U- ¹⁵ N]; 2.9–3.4 mM; pH 6.4; 298 K; <i>I</i> = 220 mM. PDB: pH 7.0; <i>I</i> = 170 mM; 20 mM phosphate buffer, 50 mM NaCl. BMRB/PDB: no D ₂ O fraction listed.	HIS47→HIE; HIS100→HIE; HIS105→HIP; HIS109→HIE; <i>Retained</i> : CYS8, CYS20, CYS34, CYS53, and CYS71 free thiols; <i>Termini</i> : NMET1/CALA135; <i>OXT</i> : present.	<i>Proxy</i> <i>Achieved</i> <i>Final I</i> <i>R</i> <i>HH</i> <i>Target</i>	phosphate as 25.36 mM NaCl <i>I</i> _{bulk} = 75.23 mM (genion : 53 NA, 40 CL) <i>I</i> = 87.46 mM after neutralisation $pH_{\text{eff}} = 6.40$ 42 titratable; 1 non-default; 3 marginal <i>I</i> = 75.36 mM, from NaCl 50.00 and phosphate 25.36	Solute charge −13; ions 53 Na ⁺ , 40 Cl ⁻ ; water 28 534; net charge 0.
BMRB 5038; PDB 1JAS	BMRB: Ubc2b [U- ¹³ C], [U- ¹⁵ N]; 1.0 mM; pH 6.7; 308 K; no D ₂ O fraction listed; 75 mM [U- ² H] ammonium acetate, 100 mM ammonium sulfate.	HIS78→HIE; <i>Retained</i> : CYS88 free thiol; <i>Termini</i> : NMET1/CSER152; <i>OXT</i> : present.	<i>Proxy</i> <i>Achieved</i> <i>Final I</i> <i>R</i> <i>HH</i> <i>Target</i>	sulfate and acetate as 274.15 mM NaCl <i>I</i> _{bulk} = 374.75 mM (genion : 144 NH ₄ ; 198 NA, 337 CL) <i>Final I</i> <i>I</i> = 471.21 mM after neutralisation $pH_{\text{eff}} = 6.70$ 43 titratable; 0 non-default; 0 marginal <i>I</i> = 374.15 mM, from ammonium acetate 74.15 and ammonium sulfate 300.00	Solute charge −5; ions 144 NH ₄ ⁺ , 198 Na ⁺ , 337 Cl ⁻ ; water 37 508; net charge 0.
BMRB 5053; PDB 1IFW	BMRB: Pab1p/PABC C-terminal domain [U- ¹⁵ N]; 3 mM; pH 6.3; 303 K; no D ₂ O fraction listed; <i>I</i> = 150 mM; 150 mM NaCl. PDB: 50 mM phosphate buffer, 100 mM NaCl, 1 mM NaN ₃ ; <i>I</i> = 200 mM; solvent metadata includes D ₂ O samples.	HIS70→HIE; <i>Termini</i> : NGLY1/CALA92; <i>OXT</i> : present.	<i>Proxy</i> <i>Achieved</i> <i>Final I</i> <i>R</i> <i>HH</i> <i>Target</i>	phosphate as 60.95 mM NaCl <i>I</i> _{bulk} = 210.58 mM (genion : 174 NA, 169 CL) <i>Final I</i> <i>I</i> = 213.69 mM after neutralisation $pH_{\text{eff}} = 6.30$ 31 titratable; 0 non-default; 1 marginal <i>I</i> = 210.95 mM, from NaCl 150.00 and phosphate 60.95	Solute charge −5; ions 174 Na ⁺ , 169 Cl ⁻ ; water 42 470; net charge 0.

Record	Provided conditions	Protein preparation	R calculations		GROMACS
BMRB 5105; PDB 1JYG	BMRB: EC298 [U- ¹³ C], [U- ¹⁵ N]; 1.0 mM; pH 6.5; 298 K; no D ₂ O fraction listed; no solution components listed.	<i>Termini</i> : NMET1/CTRP69; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Ion plan</i>	$pH_{\text{eff}} = 6.50$ 33 titratable; 0 non-default; 0 marginal neutralisation only (genion -conc 0); no salt target from listed components	Solute charge -1; ions 1 Na ⁺ ; water 21 038; net charge 0.
BMRB 5141; PDB 1T4Z	BMRB: SasA N-terminal domain [U- ¹³ C], [U- ¹⁵ N]; 0.6 mM; pH 7.0; 298 K; <i>I</i> = 220 mM. PDB: 95% H ₂ O/5% D ₂ O; 20 mM phosphate buffer, 100 mM NaCl.	HIS27→HIE; <i>Termini</i> : NGLY1/CPHE105; HIS52→HIE; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 7.00$ 17 titratable; 0 non-default; 0 marginal <i>I</i> = 135.26 mM, from NaCl 100.00 and phosphate 35.26 phosphate as 35.26 mM NaCl <i>I</i> _{bulk} = 134.98 mM (genion : 63 NA, 62 CL) <i>I</i> = 136.06 mM after neutralisation	Solute charge -1; ions 63 Na ⁺ , 62 Cl ⁻ ; water 24 083; net charge 0.
BMRB 5174; PDB 1IRZ	BMRB: ARR10-B DNA-binding motif [U- ¹³ C], [U- ¹⁵ N]; 1 mM; pH 6.9; 288 K; no D ₂ O fraction listed. PDB: 50 mM sodium phosphate buffer, 500 mM NaCl; <i>OXT</i> : present. <i>I</i> = 500 mM.	HIS12→HIE; HIS15→HIE; HIS24→HIE; HIS52→HIE; <i>Termini</i> : NTHR1/CSER64; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.90$ 24 titratable; 0 non-default; 0 marginal <i>I</i> = 582.88 mM, from NaCl 500.00 and phosphate 82.88 phosphate as 82.88 mM NaCl <i>I</i> _{bulk} = 583.66 mM (genion : 225 NA, 232 CL) <i>I</i> = 592.74 mM after neutralisation	Solute charge +7; ions 225 Na ⁺ , 232 Cl ⁻ ; water 20 122; net charge 0.
BMRB 5187; PDB 1V4R	BMRB: TraR repressor fragment [U- ¹³ C], [U- ¹⁵ N]; 1.0 mM; pH 6.5; 303 K. PDB: 90% H ₂ O/10% D ₂ O; 50 mM phosphate buffer, 100 mM NaCl; <i>I</i> = 100 mM.	HIS19→HIE; <i>Termini</i> : NGLY1/CGLU102; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.50$ 30 titratable; 0 non-default; 0 marginal <i>I</i> = 166.31 mM, from NaCl 100.00 and phosphate 66.31 phosphate as 66.31 mM NaCl <i>I</i> _{bulk} = 166.01 mM (genion : 131 NA, 136 CL) <i>I</i> = 169.18 mM after neutralisation	Solute charge +5; ions 131 Na ⁺ , 136 Cl ⁻ ; water 41 872; net charge 0.
BMRB 5225; PDB 1JNS	BMRB/PDB: Parvulin 10 [U- ¹⁵ N]; 0.8 mM; pH 6.0; 297.6 K; no D ₂ O fraction listed; 10 mM phosphate buffer, 100 mM KCl; <i>I</i> = 130 mM. PDB: 1 mM EDTA, 1 mM DTE.	HIS8→HIE; HIS37→HIE; HIS78→HIP; HIS84→HIE; <i>Retained</i> : CYS40 and CYS68 free thiols; <i>Termini</i> : NALA1/CASN92; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.00$ 31 titratable; 1 non-default; 2 marginal <i>I</i> = 115.44 mM, from KCl 100.00, phosphate 11.16, and EDTA 4.28 phosphate and EDTA as 15.44 mM NaCl <i>I</i> _{bulk} = 115.16 mM (genion : 5 NA; 36 K, 46 CL) <i>I</i> = 122.19 mM after neutralisation	Solute charge +5; ions 36 K ⁺ , 5 Na ⁺ , 46 Cl ⁻ ; water 18 697; net charge 0.
BMRB 5238; PDB 1KMX, 1VGH, 2VGH	BMRB/PDB: VEGF-165 heparin-binding domain; 1 mM; pH 5.5; 300 K; no D ₂ O fraction listed. PDB: 25 mM sodium acetate, 50 mM NaCl.	HIS16→HIP; <i>Bonded</i> : CYS7–CYS25, CYS10–CYS27, CYS29–CYS48, and CYS36–CYS50 S–S; residues CYX; <i>Termini</i> : NALA1/CARG55; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 5.50$ 31 titratable; 9 non-default; 0 marginal <i>I</i> = 71.15 mM, from NaCl 50.00 and sodium acetate 21.15 sodium acetate as 21.15 mM NaCl <i>I</i> _{bulk} = 70.38 mM (genion : 45 NA, 53 CL) <i>I</i> = 76.64 mM after neutralisation	Solute charge +8; ions 45 Na ⁺ , 53 Cl ⁻ ; water 34 354; net charge 0.
BMRB 5276; PDB 1KMA	BMRB/PDB: DIPETALIN domain I [U- ¹⁵ N]; 1.0 mM; pH 6.68; 288 K; 6% D ₂ O; 50 mM phosphate buffer, 100 mM NaCl, 0.1 mM EDTA.	HIS13→HIE; HIS34→HIE; HIS44→HIP; HIS51→HIE; HIS53→HIP; <i>Bonded</i> : CYS6–CYS31, CYS8–CYS27, and CYS16–CYS48 S–S; residues CYX; <i>Termini</i> : NPHE1/CPHE55; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.68$ 25 titratable; 8 non-default; 3 marginal <i>I</i> = 173.32 mM, from NaCl 100.00, phosphate 72.79, and EDTA 0.53 phosphate and EDTA as 73.32 mM NaCl <i>I</i> _{bulk} = 173.89 mM (genion : 80 NA, 76 CL) <i>I</i> = 178.46 mM after neutralisation	Solute charge -4; ions 80 Na ⁺ , 76 Cl ⁻ ; water 23 331; net charge 0.
BMRB 5311; PDB 1R5E	BMRB: AvrPto [U- ¹⁵ N], 1.0 mM; pH 6.0; 298 K. PDB: 10 mM Na ₂ HPO ₄ , 5 mM KH ₂ PO ₄ , 225 mM NaCl; 93% H ₂ O/7% D ₂ O in two mixtures and one 100% D ₂ O alternative.	HIS13→HIP; HIS26→HIE; HIS47→HIP; HIS59→HIP; HIS75→HIP; HIS97→HIP; HIS102→HIP; <i>Termini</i> : NASP1/CILE105; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.00$ 31 titratable; 6 non-default; 4 marginal <i>I</i> = 260.00 mM, from NaCl 225.00, Na ₂ HPO ₄ 30.00, and KH ₂ PO ₄ 5.00 phosphate as 35.00 mM NaCl <i>I</i> _{bulk} = 259.94 mM (genion : 153 NA, 154 CL) <i>I</i> = 260.79 mM after neutralisation	Solute charge +1; ions 153 Na ⁺ , 154 Cl ⁻ ; water 31 052; net charge 0.

Record	Provided conditions	Protein preparation	R calculations		GROMACS
BMRB 5328; PDB 1L7B	BMRB: DNA-ligase BRCT [U-99% ^{13}C ; U-99% ^{15}N], 1.3 mM; pH 6.2; 293 K. PDB: 300 mM ammonium acetate; 95% H_2O and 10% D_2O ; stated $I = 300$ mM ammonium acetate.	<i>Termini</i> : NMET1/CSER92; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.20$ 34 titratable; 0 non-default; 0 marginal $I = 289.49$ mM, from acetate ammonium acetate as 289.49 mM NaCl $I_{\text{bulk}} = 290.41$ mM (genion : 103 NA, 106 CL) $I = 294.64$ mM after neutralisation	Solute charge +3; ions 103 Na^+ , 106 Cl^- ; water 18 441; net charge 0.
BMRB 5329; PDB 1L7Y	BMRB: <i>C. elegans</i> ZK652.3 [U-99% ^{15}N], 0.9 mM; pH 5.5; 298 K; 10 mM ammonium acetate, 50 mM NaCl; $I = 0.06$ M. PDB: 10 mM DTT; 5% D_2O in the selected mixture.	HIS80→HIP; <i>Termini</i> : NMET1/CHIP94; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 5.50$ 20 titratable; 2 non-default; 0 marginal $I = 58.46$ mM, from NaCl 50.00 and acetate 8.46 ammonium acetate as 8.46 mM NaCl $I_{\text{bulk}} = 58.97$ mM (genion : 50 NA, 55 CL) $I = 61.92$ mM after neutralisation	Solute charge +5; ions 50 Na^+ , 55 Cl^- ; water 45 306; net charge 0.
BMRB 5375; PDB 1LD6	BMRB: BPTI.8A, 3 mM protein; pH 2.9; 298 K; 90% H_2O /10% D_2O ; $I = 0$ M; no additional solution components listed.	ASP3→ASH; ASP50→ASH; GLU7→GLH; GLU49→GLH; <i>Bonded</i> : CYS5–CYS55, CYS14–CYS38, and CYS30–CYS51 S–S; residues CYX; <i>Termini</i> : NARG1/CALA58; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Ion plan</i>	$pH_{\text{eff}} = 2.90$ 23 titratable; 10 non-default; 1 marginal neutralisation only (genion -conc 0); no salt target from listed components	Solute charge +8; ions 8 Cl^- ; water 18 952; net charge 0.
BMRB 5499; PDB 1M5Z	BMRB: GRIP1 PDZ7 domain, 1.0 mM; pH 6.5; 298 K; 99.9% D_2O ; 100 mM potassium phosphate buffer; $I = 100$ mM.	HIS8→HIE; HIS61→HIE; <i>Retained</i> : CYS69 and CYS70 free thiols; <i>Termini</i> : NSER1/CPRO91; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.50$ 31 titratable; 0 non-default; 2 marginal $I = 132.63$ mM, from phosphate potassium phosphate as 132.63 mM NaCl $I_{\text{bulk}} = 131.81$ mM (genion : 55 NA, 54 CL) $I = 133.03$ mM after neutralisation	Solute charge −1; ions 55 Na^+ , 54 Cl^- ; water 21 520; net charge 0.
BMRB 5537; PDB 1H40	BMRB: Human FF domain [U-99% ^{13}C ; U-99% ^{15}N], 2.0 mM; pH 6.0; 298 K; no D_2O fraction listed; 50 mM potassium phosphate, 150 mM NaCl; $I = 0.2$ M.	<i>Termini</i> : NGLY1/CLYS71; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.00$ 27 titratable; 0 non-default; 0 marginal $I = 205.80$ mM, from NaCl 150.00 and phosphate 55.80 potassium phosphate as 55.80 mM NaCl $I_{\text{bulk}} = 205.46$ mM (genion : 100 NA, 107 CL) $I = 212.65$ mM after neutralisation	Solute charge +7; ions 100 Na^+ , 107 Cl^- ; water 25 936; net charge 0.
BMRB 5563; PDB 1P94	BMRB: ParG subunit [U-95% ^{15}N], 1–2 mM; pH 5.5; 293 K. PDB: 100 mM NaCl, 50 mM phosphate buffer, 1 mM DTT; 100% D_2O and 90% H_2O /10% D_2O variants; stated $I = 150$ mM.	HIS7→HIP; HIS46→HIP; <i>Retained</i> : CYS53 free thiol; <i>Termini</i> : NMET1/CGLU76; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 5.50$ 29 titratable; 2 non-default; 0 marginal $I = 151.89$ mM, from NaCl 100.00 and phosphate 51.89 phosphate buffer as 51.89 mM NaCl $I_{\text{bulk}} = 152.11$ mM (genion : 156 NA, 161 CL) $I = 154.55$ mM after neutralisation	Solute charge +5; ions 156 Na^+ , 161 Cl^- ; water 54 950; net charge 0.
BMRB 5689; PDB 1QWV, 1TWO, 2JPO	BMRB: pheromone-binding protein [U-99% ^{13}C ; U-99% ^{15}N], 1.0 mM; pH 6.33; 308 K. PDB: 50 mM phosphate buffer, 1 mM EDTA, 0.1% NaN_3 , 95% H_2O /5% D_2O ; stated $I = 50$ mM phosphate.	HIS69→HIP; HIS70→HIE; HIS80→HIE; HIS95→HIE; HIS123→HIE; <i>Bonded</i> : CYS19–CYS54, CYS50–CYS108, and CYS97–CYS117 S–S; residues CYX; <i>Termini</i> : NSER1/CVAL142; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.33$ 52 titratable; 7 non-default; 3 marginal $I = 66.48$ mM, from phosphate 61.64 and EDTA 4.84 phosphate and EDTA as 66.48 mM NaCl $I_{\text{bulk}} = 67.60$ mM (genion : 34 NA, 25 CL) $I = 79.77$ mM after neutralisation	Solute charge −9; ions 34 Na^+ , 25 Cl^- ; water 19 408; net charge 0.
BMRB 5691; PDB 1NY4	BMRB: 30S ribosomal protein S28E (RS28.PYRHO/JR19) [U-100% ^{13}C ; U-100% ^{15}N], 1.0 mM; pH 6.5; 293 K; 5% D_2O ; 20 mM MES, 100 mM NaCl, 5 mM CaCl_2 , 10 mM DTT, 0.02% NaN_3 . PDB: stated $I = 100$ mM NaCl.	HIS77→HIE; HIS78→HIE; HIS79→HIP; HIS80→HIP; HIS81→HIE; <i>Termini</i> : NMET1/CHIE82; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.50$ 38 titratable; 2 non-default; 5 marginal $I = 120.69$ mM, from NaCl 100.00, MES 5.69, and CaCl_2 15.00 MES as 5.69 mM NaCl $I_{\text{bulk}} = 119.46$ mM (genion : 2 CA; 47 NA, 55 CL) $I = 123.97$ mM after neutralisation	Solute charge +4; ions 2 Ca^{2+} , 47 Na^+ , 55 Cl^- ; water 23 483; net charge 0.

Record	Provided conditions	Protein preparation	R calculations		GROMACS
BMRB 5763; PDB 1RQ8	BMRB: SAV1595 from <i>Staphylococcus aureus</i> [U- ¹³ C; U- ¹⁵ N], 0.3–1 mM; pH 6.5; 298 K; 75 mM potassium phosphate buffer; <i>I</i> = 0.45 M. PDB: pH 7.5; 75 mM phosphate buffer; 8% D ₂ O; stated <i>I</i> = 0.45.	HIS15→HIE; HIS49→HIE; <i>Termini</i> : NMET1/CPRO96; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.50$ 35 titratable; 0 non-default; 1 marginal <i>I</i> = 75.00 mM, from KH ₂ PO ₄ 75.00 phosphate as 75.00 mM NaCl <i>I</i> _{bulk} = 75.43 mM (genion : 32 NA, 30 CL) <i>I</i> = 77.95 mM after neutralisation	Solute charge −2; ions 32 Na ⁺ , 30 Cl [−] ; water 21 005; net charge 0.
BMRB 5801; PDB 1P9J	BMRB: T1E [U- ¹⁵ N]; 0.8 mM; pH 6.3; 298 K; 50 mM K ₂ HPO ₄ /KH ₂ PO ₄ . PDB: 95% H ₂ O/5% D ₂ O; 50 mM phosphate buffer; <i>I</i> = 50 mM.	HIS4→HIP; HIS12→HIE; HIS18→HIP; <i>Bonded</i> : CYS8–CYS22, CYS16–CYS33, and CYS35–CYS44 S–S; residues CYX; <i>Termini</i> : NVAL1/CLEU54; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.30$ 27 titratable; 8 non-default; 1 marginal <i>I</i> = 60.95 mM, from phosphate 60.95 phosphate as 60.95 mM NaCl <i>I</i> _{bulk} = 62.05 mM (genion : 26 NA, 24 CL) <i>I</i> = 64.64 mM after neutralisation	Solute charge −2; ions 26 Na ⁺ , 24 Cl [−] ; water 20 692; net charge 0.
BMRB 5846; PDB 1PUZ	BMRB: Hypothetical protein NMA1147 [U- ¹³ C], [U- ¹⁵ N]; 1.0 mM; pH 6.5; 298 K. PDB: 90% H ₂ O/10% D ₂ O; 20 mM MES, 100 mM NaCl, 5 mM CaCl ₂ , 10 mM DTT, 0.02% azide.	HIS37→HIP; HIS64→HIE; HIS71→HIP; <i>Termini</i> : NMET1/CALA82; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.50$ 34 titratable; 2 non-default; 2 marginal <i>I</i> = 120.69 mM, from NaCl 100.00, CaCl ₂ 15.00, and MES 5.69 MES as 5.69 mM NaCl <i>I</i> _{bulk} = 124.94 mM (genion : 2 CA; 34 NA, 36 CL) <i>I</i> = 126.56 mM after neutralisation	Solute charge −2; ions 2 Ca ²⁺ , 34 Na ⁺ , 36 Cl [−] ; water 16 446; net charge 0.
BMRB 5847; PDB 1PV0, 3FYR	BMRB: Sda [U- ¹³ C], [U- ¹⁵ N]; 1.0 mM; pH 6.85; 298 K; selected condition is 100% H ₂ O; <i>I</i> = 0.125 M. PDB: 20 mM sodium phosphate, 100 mM NaCl.	HIS40→HIE; <i>Termini</i> : NMET1/CSER46; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.85$ 18 titratable; 0 non-default; 0 marginal <i>I</i> = 132.15 mM, from NaCl 100.00 and phosphate 32.15 phosphate as 32.15 mM NaCl <i>I</i> _{bulk} = 132.78 mM (genion : 40 NA, 39 CL) <i>I</i> = 134.48 mM after neutralisation	Solute charge −1; ions 40 Na ⁺ , 39 Cl [−] ; water 15 546; net charge 0.
BMRB 5849; PDB 1UFM	BMRB/PDB: COP9 complex subunit 4 [U- ¹³ C], [U- ¹⁵ N]; 6.1 mM; pH 6.0; 298 K; 90% H ₂ O/10% D ₂ O; 20 mM sodium phosphate buffer, 100 mM NaCl, 1 mM d-DTT, 1 mM [U- ² H] dithiothreitol; <i>I</i> = 0.12 M.	HIS19→HIP; HIS72→HIP; <i>Termini</i> : NGLY1/CGLY84; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.00$ 21 titratable; 2 non-default; 2 marginal <i>I</i> = 122.32 mM, from NaCl 100.00 and phosphate 22.32 phosphate as 22.32 mM NaCl <i>I</i> _{bulk} = 121.90 mM (genion : 87 NA, 84 CL) <i>I</i> = 124.08 mM after neutralisation	Solute charge −3; ions 87 Na ⁺ , 84 Cl [−] ; water 36 908; net charge 0.
BMRB 5873; PDB 1P6R, 2P7C	BMRB/PDB: BlaI N-terminal domain [U- ¹³ C], [U- ¹⁵ N]; 0.5 mM; pH 7.6; 298 K; 90% H ₂ O/10% D ₂ O; 75 mM phosphate buffer, 300 mM KCl; <i>I</i> = 375 mM.	HIS21→HIE; HIS59→HIE; HIS60→HIE; <i>Termini</i> : NMET1/CSER82; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 7.60$ 29 titratable; 0 non-default; 1 marginal <i>I</i> = 481.58 mM, from KCl 300.00 and phosphate 181.58 phosphate as 181.58 mM NaCl <i>I</i> _{bulk} = 480.58 mM (genion : 76 NA; 125 K, 204 CL) <i>I</i> = 484.17 mM after neutralisation	Solute charge +3; ions 125 K ⁺ , 76 Na ⁺ , 204 Cl [−] ; water 21 932; net charge 0.
BMRB 5875; PDB 1Q10	BMRB: Immunoglobulin Binding domain 1 of Protein G [U-95% ¹³ C], [U-95% ¹⁵ N]; 1.7 mM; pH 6.0; 298 K. PDB: pH 5.5; 92.5% H ₂ O/7.5% D ₂ O; 50 mM phosphate buffer; 0.02% NaN ₃ ; <i>I</i> = 50 mM sodium phosphate buffer.	<i>Termini</i> : NMET1/CGLU56; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.00$ 19 titratable; 0 non-default; 0 marginal <i>I</i> = 55.80 mM, from phosphate 55.80 phosphate as 55.80 mM NaCl <i>I</i> _{bulk} = 56.05 mM (genion : 18 NA, 14 CL) <i>I</i> = 64.06 mM after neutralisation	Solute charge −4; ions 18 Na ⁺ , 14 Cl [−] ; water 13 274; net charge 0.
BMRB 5963; PDB 2APN	BMRB: hypothetical protein hi1723 [U-95% ¹³ C], [U-90% ¹⁵ N]; 1.0 mM; pH 7.0; 298 K. PDB: 90% H ₂ O/10% D ₂ O; 50 mM potassium phosphate, 100 mM NaCl, 4 mM DTT; <i>I</i> = 50 – −100 mM.	<i>Bonded</i> : CYS42–CYS106 S–S; residues CYX; <i>Retained</i> : CYS108 free thiol; <i>Termini</i> : NMET1/CILE114; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 7.00$ 31 titratable; 2 non-default; 0 marginal <i>I</i> = 188.14 mM, from NaCl 100.00 and phosphate 88.14 phosphate as 88.14 mM NaCl <i>I</i> _{bulk} = 189.08 mM (genion : 98 NA, 89 CL) <i>I</i> = 198.64 mM after neutralisation	Solute charge −9; ions 98 Na ⁺ , 89 Cl [−] ; water 24 562; net charge 0.
BMRB 5977; PDB 1R73	BMRB/PDB: TM1492 [U- ¹³ C], [U- ¹⁵ N]; 3.6 mM; pH* 6.0; 313 K; 90% H ₂ O/10% D ₂ O; 20 mM phosphate buffer. PDB: <i>I</i> not listed.	<i>Termini</i> : NMET1/CARG66; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.04$ 29 titratable; 0 non-default; 0 marginal <i>I</i> = 22.32 mM, from phosphate 22.32 phosphate as 22.32 mM NaCl <i>I</i> _{bulk} = 22.06 mM (genion : 9 NA, 16 CL) <i>I</i> = 30.65 mM after neutralisation	Solute charge +7; ions 9 Na ⁺ , 16 Cl [−] ; water 21 696; net charge 0.

Record	Provided conditions	Protein preparation	R calculations		GROMACS
BMRB 6010; PDB 1RHX, 1X9A	BMRB: TM0979 [U- ¹³ C], [U- ¹⁵ N]; 2–10 mM; pH* 6.0; 313 K. PDB: pH 6.5; 300 K; 95% H ₂ O/5% D ₂ O; 25 mM sodium phosphate, 450 mM NaCl; <i>I</i> not listed.	HIS12→HIP; <i>Termini</i> : NMET1/CGLY87; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.02$ 34 titratable; 1 non-default; 1 marginal $I = 477.90$ mM, from NaCl 450.00 and phosphate 27.90 phosphate as 27.90 mM NaCl $I_{\text{bulk}} = 476.70$ mM (genion : 154 NA, 149 CL) $I = 484.70$ mM after neutralisation	Solute charge −5; ions 154 Na ⁺ , 149 Cl [−] ; water 16 223; net charge 0.
BMRB 6028; PDB 1RQ6	BMRB: ribosomal protein S17E [U- ¹³ C; U- ¹⁵ N], 1.0 mM; pH 6.5; 298 K; 450 mM NaCl; $I = 0.45$ M. PDB: NMET1/CLYS62; 25 mM Na ₂ PO ₄ ; 95% H ₂ O/5% D ₂ O. <i>OXT</i> : present.	HIS20→HIE; HIS45→HIE; <i>Termini</i> : NMET1/CLYS62; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.50$ 23 titratable; 0 non-default; 0 marginal $I = 483.16$ mM, from NaCl 450.00 and phosphate 33.16 phosphate as 33.16 mM NaCl $I_{\text{bulk}} = 481.78$ mM (genion : 154 NA, 159 CL) $I = 489.60$ mM after neutralisation	Solute charge +5; ions 154 Na ⁺ , 159 Cl [−] ; water 16 784; net charge 0.
BMRB 6031; PDB 1TOZ	BMRB: human Notch-1 [U- ¹⁵ N], 1.0 mM; pH 6.1; 298 K; no D ₂ O fraction listed; 15 mM CaCl ₂ , 0.02% NaN ₃ ; $I = 0.09$ M. PDB: 15 mM CaCl ₂ in the ionic-strength field.	HIS17→HIP; HIS78→HIP; HIS93→HIE; HIS115→HIP; <i>Bonded</i> : CYS8–CYS21, CYS15–CYS30, CYS32–CYS41, CYS48–CYS59, CYS53–CYS68, CYS70–CYS79, CYS86–CYS97, CYS91–CYS106, and CYS108–CYS117 S–S; residues CYX; <i>Termini</i> : NSER1/CGLN118; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.10$ 47 titratable; 21 non-default; 3 marginal $I = 45.00$ mM, from CaCl ₂ 45.00 $I_{\text{bulk}} = 45.27$ mM (genion : 32 CA, 53 CL) $I = 52.53$ mM after neutralisation	Solute charge −11; ions 32 Ca ²⁺ , 53 Cl [−] ; water 92 993; net charge 0.
BMRB 6122; PDB 1SLJ, 1SMX, 1SN8, 2BX2, 2C0B, 2C4R, 2VMK, 2VRT	BMRB: S1 RNase E [U- ¹³ C; U- ¹⁵ N], 1.5 mM; pH 6.5; 303 K; no D ₂ O fraction listed; no solution components listed.	HIS3→HIE; HIS36→HIE; HIS56→HIE; <i>Termini</i> : NGLY1/CSER96; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Ion plan</i>	$pH_{\text{eff}} = 6.50$ 35 titratable; 0 non-default; 3 marginal neutralisation only (genion -conc 0); no salt target from listed components	Solute charge +1; ions 1 Cl [−] ; water 24 317; net charge 0.
BMRB 6167; PDB 1SNL	BMRB: nucleobindin [U- ¹⁵ N], 1.0 mM; pH 6.8; 300 K; 90% H ₂ O/10% D ₂ O; 5 mM CaCl ₂ , 0.1 mM NaN ₃ ; $I = 10$ mM. PDB: 5 mM CaCl ₂ , 0.1 mM NaN ₃ , 5 mM KCl.	HIS25→HIE; HIS72→HIE; <i>Termini</i> : NLEU1/CPHE99; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.80$ 46 titratable; 0 non-default; 1 marginal $I = 20.00$ mM, from KCl 5.00 and CaCl ₂ 15.00 $I_{\text{bulk}} = 19.00$ mM (genion : 3 K; 9 CA, 7 CL) $I = 51.42$ mM after neutralisation	Solute charge −14; ions 9 Ca ²⁺ , 3 K ⁺ , 7 Cl [−] ; water 23 716; net charge 0.
BMRB 6175; PDB 1SZL	BMRB: F-spondin TSR domain 1 [U- ¹³ C; U- ¹⁵ N], 1.0 mM; pH 6.5; 283 K; no D ₂ O fraction listed; no solution components listed. PDB: 20 mM Bis-Tris buffer.	<i>Bonded</i> : CYS5–CYS42, CYS16–CYS51, and CYS21–CYS56 S–S; residues CYX; <i>Termini</i> : NGLY1/CGLY61; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.50$ 22 titratable; 6 non-default; 0 marginal $I = 9.54$ mM, from Bis-Tris 9.54 Bis-Tris as 9.54 mM NaCl $I_{\text{bulk}} = 8.49$ mM (genion : 6 NA, 4 CL) $I = 10.61$ mM after neutralisation	Solute charge −2; ions 6 Na ⁺ , 4 Cl [−] ; water 25 013; net charge 0.
BMRB 6188; PDB 1ST7	BMRB: yeast acyl-CoA-binding protein [U- ¹³ C; U- ¹⁵ N], 2.5 mM; pH 6.75; 298 K; 90% H ₂ O/10% D ₂ O; no solution components listed. PDB: <i>I</i> NULL.	<i>Termini</i> : NVAL1/CSER86; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Ion plan</i>	$pH_{\text{eff}} = 6.75$ 36 titratable; 0 non-default; 0 marginal neutralisation only (genion -conc 0); no salt target from listed components	Solute charge −4; ions 4 Na ⁺ ; water 22 234; net charge 0.
BMRB 6189; PDB 1SS6	BMRB: NSFL1 cofactor p47 [U- ¹⁵ N], 2 mM; pH 5.6; 300 K; 90% H ₂ O/10% D ₂ O; 20 mM phosphate buffer, 100 mM sodium chloride; $I = 120$ mM. PDB: 20 mM phosphate buffer Na, 150 mM sodium chloride.	HIS7→HIP; HIS13→HIP; HIS59→HIP; HIS70→HIP; <i>Termini</i> : NGLY1/CTHR102; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 5.60$ 35 titratable; 4 non-default; 0 marginal $I = 120.95$ mM, from NaCl 100.00 and phosphate 20.95 phosphate as 20.95 mM NaCl $I_{\text{bulk}} = 120.90$ mM (genion : 78 NA, 81 CL) $I = 123.23$ mM after neutralisation	Solute charge +3; ions 78 Na ⁺ , 81 Cl [−] ; water 34 249; net charge 0.
BMRB 6197; PDB 1RUW, 1VA7, 2BTT	BMRB: Src Homology-3 domain [U-95% ¹³ C; U-90% ¹⁵ N], 0.6 mM; pH 6.9; 298 K; no D ₂ O fraction listed; $I = 0.1$ M; no solution components listed.	<i>Termini</i> : NGLY1/CVAL70; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.90$ 23 titratable; 0 non-default; 0 marginal $I = 100.00$ mM, from a deposited scalar value deposited ionic-strength scalar as 100.00 mM NaCl $I_{\text{bulk}} = 99.75$ mM (genion : 45 NA, 44 CL) $I = 100.88$ mM after neutralisation	Solute charge −1; ions 45 Na ⁺ , 44 Cl [−] ; water 23 473; net charge 0.

Record	Provided conditions	Protein preparation	R calculations		GROMACS
BMRB 6208; PDB 1TUZ	BMRB: diacylglycerol kinase alpha (HR532) [U- ¹³ C; U- ¹⁵ N], 1.1 mM; pH 6.5; 298 K; 95% H ₂ O/5% D ₂ O; 100 mM NaCl, 5 mM CaCl ₂ , 20 mM MES, 10 mM DTT, 0.02% NaN ₃ .	HIS69→HIE; HIS82→HIP; <i>Bonded</i> : CYS83–CYS95 S–S; residues CYX; <i>Retained</i> : CYS101 free thiol; <i>Termini</i> : NMET1/CSER118; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.50$ 43 titratable; 3 non-default; 2 marginal $I = 120.69$ mM, from NaCl 100.00, CaCl ₂ 15.00, and MES 5.69 MES as 5.69 mM NaCl $I_{\text{bulk}} = 119.65$ mM (genion : 2 CA; 59 NA, 57 CL) $I = 126.80$ mM after neutralisation	Solute charge −6; ions 2 Ca ²⁺ , 59 Na ⁺ , 57 Cl [−] ; water 25 825; net charge 0.
BMRB 6209; PDB 1TIZ	BMRB: N-terminal domain of At3g03410.1 [U- ¹⁵ N], 1 mM; pH 6.4; 298 K; no D ₂ O fraction listed; 5 mM CaCl ₂ , 20 mM Bis-Tris, 5 mM DTT, 1 mM DSS.	<i>Retained</i> : CYS62 free thiol; <i>Termini</i> : NSER1/CLEU67; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.40$ 28 titratable; 0 non-default; 0 marginal $I = 25.69$ mM, from Bis-Tris 10.69 and CaCl ₂ 15.00 Bis-Tris as 10.69 mM NaCl $I_{\text{bulk}} = 23.30$ mM (genion : 1 CA; 10 NA, 5 CL) $I = 36.90$ mM after neutralisation	Solute charge −7; ions 1 Ca ²⁺ , 10 Na ⁺ , 5 Cl [−] ; water 13 239; net charge 0.
BMRB 6254; PDB 1TXE	BMRB: HPr I14A mutant from <i>S. carnosus</i> [U- ¹⁵ N], 1.7 mM; pH 7.14; 298 K; no D ₂ O fraction listed; 20 mM potassium phosphate, 100 mM KCl. PDB: $I = 0.12$ M.	HIS15→HIE; <i>Termini</i> : NMET1/CGLU88; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 7.14$ 27 titratable; 0 non-default; 0 marginal $I = 138.39$ mM, from KCl 100.00 and phosphate 38.39 phosphate as 38.39 mM NaCl $I_{\text{bulk}} = 138.27$ mM (genion : 14 NA; 43 K, 49 CL) $I = 149.55$ mM after neutralisation	Solute charge −8; ions 43 K ⁺ , 14 Na ⁺ , 49 Cl [−] ; water 18 549; net charge 0.
BMRB 6258; PDB 1S62	BMRB: TolA domain III [U- ¹³ C; U- ¹⁵ N], 0.4 mM; pH 6.8; 300 K; 30 mM potassium phosphate buffer, 30 mM NaCl; $I = 60$ mM. PDB: 90% H ₂ O/10% D ₂ O.	HIS101→HIE; HIS102→HIE; HIS103→HIE; HIS104→HIE; HIS105→HIE; <i>Bonded</i> : CYS42–CYS67 S–S; residues CYX; <i>Termini</i> : NALA1/CHIE106; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.80$ 35 titratable; 2 non-default; 1 marginal $I = 76.80$ mM, from NaCl 30.00 and phosphate 46.80 phosphate as 46.80 mM NaCl $I_{\text{bulk}} = 76.93$ mM (genion : 48 NA, 50 CL) $I = 78.53$ mM after neutralisation	Solute charge +2; ions 48 Na ⁺ , 50 Cl [−] ; water 33 302; net charge 0.
BMRB 6269; PDB 1U3M	BMRB/PDB: chicken prion protein fragment chPrP (128-242) [U- ¹³ C; U- ¹⁵ N], 1.0 mM; pH 4.3; 293 K; 95% H ₂ O/5% D ₂ O; $I = 10$ mM; 10 mM sodium acetate.	HIS22→HIP; GLU34→GLH; GLU68→GLH; GLU92→GLH; GLU94→GLH; GLU104→GLH; GLU112→GLH; <i>Bonded</i> : CYS61–CYS106 S–S; residues CYX; <i>Termini</i> : NGLY1/CSER117; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 4.30$ 41 titratable; 9 non-default; 1 marginal $I = 10.00$ mM, from a deposited scalar value sodium acetate as 10.00 mM NaCl $I_{\text{bulk}} = 9.32$ mM (genion : 5 NA, 11 CL) $I = 14.91$ mM after neutralisation	Solute charge +6; ions 5 Na ⁺ , 11 Cl [−] ; water 28 870; net charge 0.
BMRB 6291; PDB 1X9B	BMRB: hypothetical membrane protein Ta0354 [U- ¹³ C; U- ¹⁵ N], 1 mM; pH 6.5; 298 K; 450 mM NaCl; $I = 0.45$ M. PDB: 95% H ₂ O/5% D ₂ O; 25 mM Na ₂ PO ₄ ; 1 mM benzamidine, inhibitor cocktail, 0.01% NaN ₃ .	<i>Termini</i> : NARG1/CLYS53; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.50$ 20 titratable; 0 non-default; 0 marginal $I = 483.16$ mM, from NaCl 450.00 and phosphate 33.16 phosphate as 33.16 mM NaCl $I_{\text{bulk}} = 484.64$ mM (genion : 154 NA, 160 CL) $I = 494.08$ mM after neutralisation	Solute charge 6; ions 154 Na ⁺ , 160 Cl [−] ; water 16 645; net charge 0.
BMRB 6318; PDB 1XFL	BMRB/PDB: thioredoxin h1 (<i>A. thaliana</i>) [U- ¹³ C; U- ¹⁵ N], 0.5 mM; pH 5.5; 298 K; 90% H ₂ O/10% D ₂ O; 50 mM KCl, 10 mM potassium phosphate buffer; stated $I = 50$ mM.	HIS3→HIP; HIS4→HIP; HIS5→HIP; HIS6→HIP; HIS7→HIP; HIS8→HIP; HIS22→HIP; HIS122→HIP; ASP44→ASH; <i>Bonded</i> : CYS50–CYS53 S–S; residues CYX; <i>Retained</i> : CYS21 free thiol; <i>Termini</i> : NMET1/CALA124; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 5.50$ 41 titratable; 11 non-default; 0 marginal $I = 60.38$ mM, from KCl 50.00 and phosphate 10.38 phosphate as 10.38 mM NaCl $I_{\text{bulk}} = 60.06$ mM (genion : 7 NA; 32 K, 45 CL) $I = 64.68$ mM after neutralisation	Solute charge +6; ions 32 K ⁺ , 7 Na ⁺ , 45 Cl [−] ; water 34 430; net charge 0.
BMRB 6355	BMRB: Staphylococcus epidermidis protein SE0936 [U-5% ¹³ C; U-100% ¹⁵ N], 8 mM; pH 6.5; 293 K; 95% H ₂ O/5% D ₂ O; $I = 100$ mM; 100 mM NaCl, 5 mM CaCl ₂ , 20 mM MES, 10 mM DTT, 0.02% NaN ₃ .	HIS45→HIE; HIS67→HIE; HIS83→HIE; HIS84→HIE; HIS85→HIE; HIS86→HIE; HIS87→HIP; <i>Bonded</i> : CYS48–CYS51 S–S; residues CYX; <i>Retained</i> : CYS30 free thiol; <i>Termini</i> : NMET1/CHIE88; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.50$ 33 titratable; 3 non-default; 7 marginal $I = 120.69$ mM, from CaCl ₂ 15.00, NaCl 100.00, and MES 5.69 MES as 5.69 mM NaCl $I_{\text{bulk}} = 121.33$ mM (genion : 4 CA; 88 NA, 88 CL) $I = 125.92$ mM after neutralisation	Solute charge −8; ions 4 Ca ²⁺ , 88 Na ⁺ , 88 Cl [−] ; water 40 807; net charge 0.

Record	Provided conditions	Protein preparation	R calculations		GROMACS
BMRB 6368; PDB 1XN9	BMRB: hypothetical protein rps24e [U- ¹³ C; U- ¹⁵ N], 1.0 mM; pH 6.5; 298 K; 95% H ₂ O/5% D ₂ O; 100 mM NaCl, 5 mM CaCl ₂ , 20 mM MES, 10 mM DTT, 0.02% NaN ₃ .	<i>Termini</i> : NMET1/CALA101; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.50$ 42 titratable; 0 non-default; 0 marginal $I = 140.69$ mM, from NaCl 100.00, CaCl ₂ 15.00, MES 5.69, and NH ₄ OAc 20.00 MES and NH ₄ OAc as 25.69 mM NaCl $I_{\text{bulk}} = 138.93$ mM (genion : 2 CA; 59 NA, 63 CL) $I = 138.93$ mM after neutralisation	Solute charge 0; ions 2 Ca ²⁺ , 59 Na ⁺ , 63 Cl ⁻ ; water 24 476; net charge 0.
BMRB 6375; PDB 1XX3	BMRB: TonB C-terminal domain [U-95% ¹³ C; U-95% ¹⁵ N], 1.4 mM; pH 7.00; 298 K; no D ₂ O fraction listed; 100 mM sodium phosphate, 0.6 mM NaN ₃ .	<i>Termini</i> : NSER1/CGLN89; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 7.00$ 26 titratable; 0 non-default; 0 marginal $I = 176.28$ mM, from sodium phosphate sodium phosphate as 176.28 mM NaCl $I_{\text{bulk}} = 177.21$ mM (genion : 68 NA, 75 CL) $I = 186.33$ mM after neutralisation	Solute charge +7; ions 68 Na ⁺ , 75 Cl ⁻ ; water 20 276; net charge 0.
BMRB 6376	BMRB: TgMIC1-CT; 0.5 mM [U- ¹³ C; U- ¹⁵ N]; pH 7.0; 308 K; no D ₂ O fraction listed; 150 mM sodium chloride, 20 mM sodium phosphate.	HIS5→HIE; HIS36→HIE; HIS77→HIP; <i>Bonded</i> : CYS35–CYS106 S–S; residues CYX; <i>Termini</i> : NLYS1/CALA137; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 7.00$ 31 titratable; 3 non-default; 0 marginal $I = 185.26$ mM, from NaCl and phosphate phosphate as 35.26 mM NaCl $I_{\text{bulk}} = 185.32$ mM (genion : 130 NA, 122 CL) $I = 191.39$ mM after neutralisation	Solute charge −8; ions 130 Na ⁺ , 122 Cl ⁻ ; water 34 819; net charge 0.
BMRB 6378; PDB 1XYK	BMRB/PDB: pH 4.5; 293 K. PDB: 10 mM acetate; stated $I = 10$ mM. No D ₂ O fraction listed.	HIS22→HIP; HIS69→HIP; GLU78→GLH; GLU82→GLH; GLU89→GLH; <i>Bonded</i> : CYS61–CYS96 S–S; residues CYX; <i>Termini</i> : NGLY1/CSER113; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 4.50$ 43 titratable; 7 non-default; 4 marginal $I = 3.55$ mM, from acetate acetate as 3.55 mM NaCl $I_{\text{bulk}} = 3.79$ mM (genion : 2 NA, 5 CL) $I = 6.63$ mM after neutralisation	Solute charge 3; ions 2 Na ⁺ , 5 Cl ⁻ ; water 28 500; net charge 0.
BMRB 6380; PDB 1XYQ	BMRB/PDB: pH 4.5; 293 K. PDB: 10 mM acetate; stated $I = 10$ mM. No D ₂ O fraction listed.	HIS22→HIP; GLU34→GLH; HIS59→HIP; HIS69→HIP; GLU82→GLH; GLU89→GLH; GLU93→GLH; GLU103→GLH; GLU105→GLH; <i>Bonded</i> : CYS61–CYS96 S–S; residues CYX; <i>Termini</i> : NGLY1/CSER113; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 4.50$ 42 titratable; 11 non-default; 5 marginal $I = 3.55$ mM, from acetate acetate as 3.55 mM NaCl $I_{\text{bulk}} = 3.81$ mM (genion : 2 NA, 9 CL) $I = 10.49$ mM after neutralisation	Solute charge 7; ions 2 Na ⁺ , 9 Cl ⁻ ; water 27 871; net charge 0.
BMRB 6383; PDB 1XYW	BMRB/PDB: pH 4.5; 293 K. PDB: 10 mM acetate; stated $I = 10$ mM. No D ₂ O fraction listed.	HIS22→HIP; HIS59→HIP; HIS69→HIP; GLU78→GLH; GLU82→GLH; GLU89→GLH; GLU93→GLH; GLU103→GLH; <i>Bonded</i> : CYS61–CYS96 S–S; residues CYX; <i>Termini</i> : NGLY1/CSER113; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 4.50$ 42 titratable; 10 non-default; 4 marginal $I = 3.55$ mM, from acetate acetate as 3.55 mM NaCl $I_{\text{bulk}} = 3.75$ mM (genion : 2 NA, 8 CL) $I = 9.37$ mM after neutralisation	Solute charge 6; ions 2 Na ⁺ , 8 Cl ⁻ ; water 28 827; net charge 0.
BMRB 6466	BMRB: human ubiquitin; pH* 5.7; 303 K; 10% D ₂ O; 50 mM potassium phosphate.	HIS68→HIP; <i>Termini</i> : NMET1/CGLY76; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 5.74$ 25 titratable; 1 non-default; 0 marginal $I = 52.98$ mM, from phosphate phosphate as 52.98 mM NaCl $I_{\text{bulk}} = 53.43$ mM (genion : 19 NA, 20 CL) $I = 54.84$ mM after neutralisation	Solute charge 1; ions 19 Na ⁺ , 20 Cl ⁻ ; water 18 763; net charge 0.
BMRB 6469; PDB 1WXL	BMRB: pH 5.2; 298 K. PDB: 90% H ₂ O/10% D ₂ O; 2 mM sodium acetate; stated $I = 0$ mM.	HIS3→HIP; HIS64→HIP; <i>Termini</i> : NGLY1/CALA74; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 5.20$ 35 titratable; 2 non-default; 0 marginal $I = 1.47$ mM, from acetate acetate as 1.47 mM NaCl $I_{\text{bulk}} = 1.85$ mM (genion : 1 NA, 7 CL) $I = 7.38$ mM after neutralisation	Solute charge 6; ions 1 Na ⁺ , 7 Cl ⁻ ; water 29 051; net charge 0.
BMRB 6478; PDB 1YSF	BMRB: AbrB N-domain; pH 5.8; 305 K; 20.0 mM potassium phosphate, 50.0 mM KCl, 0.02% NaN ₃ . PDB: 90% H ₂ O/10% D ₂ O; stated $I = 80$ mM; 80 mM KCl.	<i>Termini</i> : NMET1/CASN53; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 5.80$ 23 titratable; 0 non-default; 0 marginal $I = 71.49$ mM, from KCl and phosphate phosphate as 21.49 mM NaCl $I_{\text{bulk}} = 72.38$ mM (genion : 7 NA; 17 K, 26 CL) $I = 75.39$ mM after neutralisation	Solute charge 2; ions 17 K ⁺ , 7 Na ⁺ , 26 Cl ⁻ ; water 17 702; net charge 0.

Record	Provided conditions	Protein preparation	R calculations		GROMACS
BMRB 6505	BMRB: Mar30; pH 6.5; 293 K; 95% H ₂ O/5% D ₂ O; 20 mM MES, 100 mM NaCl, 5 mM CaCl ₂ , 10 mM DTT; stated $I = 100$ mM.	<i>Termini</i> : NMET1/CGLU68; <i>OXT</i> : present.	<i>R</i>	$pH_{\text{eff}} = 6.50$	Solute charge -4 ; ions 2 Ca ²⁺ , 55 Na ⁺ , 55 Cl ⁻ ; water 25 244; net charge 0.
BMRB 6517; PDB 1YWW	BMRB: PA4738; 1 mM [U- ¹³ C; U- ¹⁵ N]; pH 7.8; 298 K; no D ₂ O fraction listed; 1 M NaCl, 20 mM Tris HCl. PDB: stated $I = 1$ M NaCl.	HIS35→HIE; <i>Termini</i> : NMET1/CLEU65; <i>OXT</i> : present.	<i>HH</i>	24 titratable; 0 non-default; 0 marginal	
			<i>Target</i>	$I = 120.69$ mM, from NaCl, CaCl ₂ , and MES	
			<i>Proxy</i>	MES as 5.69 mM NaCl	
			<i>Achieved</i>	$I_{\text{bulk}} = 119.07$ mM (genion : 2 CA; 55 NA, 55 CL)	
			<i>Final I</i>	$I = 123.24$ mM after neutralisation	
BMRB 6527; PDB 2B95	BMRB: pH 6.0; 315 K. PDB: 95% H ₂ O/5% D ₂ O; 50 mM MES, 0.02% NaN ₃ .	HIS47→HIP; <i>Termini</i> : NMET1/CGLU96; <i>OXT</i> : present.	<i>R</i>	$pH_{\text{eff}} = 7.80$	Solute charge -2 ; ions 306 Na ⁺ , 304 Cl ⁻ ; water 15 455; net charge 0.
			<i>HH</i>	30 titratable; 0 non-default; 1 marginal	
			<i>Target</i>	$I = 1012.91$ mM, from NaCl and Tris HCl	
			<i>Proxy</i>	Tris HCl as 12.91 mM NaCl	
			<i>Achieved</i>	$I_{\text{bulk}} = 1014.56$ mM (genion : 306 NA, 304 CL)	
BMRB 6552; PDB 1YZC	BMRB: redesigned apo-cytochrome b562; 50 mM sodium acetate; pH 4.84; 298 K; 92.5% H ₂ O/7.5% D ₂ O; no additional solution components listed. PDB: no numeric ionic strength listed.	HIS63→HIP; <i>Termini</i> : NALA1/CGLY106; <i>OXT</i> : present.	<i>Final I</i>	$I = 1017.89$ mM after neutralisation	Solute charge 1; ions 10 Na ⁺ , 11 Cl ⁻ ; water 20 055; net charge 0.
			<i>R</i>	$pH_{\text{eff}} = 6.00$	
			<i>HH</i>	28 titratable; 1 non-default; 0 marginal	
			<i>Target</i>	$I = 27.87$ mM, from MES	
			<i>Proxy</i>	MES as 27.87 mM NaCl	
BMRB 6574	BMRB: MrR5 (<i>Methanococcus maripaludis</i>), 1.1 mM [U- ¹⁵ N; U-5% ¹³ C]; pH 4.5; 293 K; 95% H ₂ O/5% D ₂ O; 5 mM CaCl ₂ , 100 mM NaCl, 20 mM NH ₄ OAc, 10 mM DTT, 0.02% NaN ₃ ; $I = 100$ mM.	GLU18→GLH; GLU20→GLH; GLU47→GLH; GLU61→GLH; GLU67→GLH; <i>Termini</i> : NMET1/CLEU70; <i>OXT</i> : present.	<i>Achieved</i>	$I_{\text{bulk}} = 26.68$ mM (genion : 10 NA, 11 CL)	Solute charge -4 ; ions 24 Na ⁺ , 20 Cl ⁻ ; water 39 498; net charge 0.
			<i>Final I</i>	$I = 28.01$ mM after neutralisation	
			<i>R</i>	$pH_{\text{eff}} = 4.84$	
			<i>HH</i>	39 titratable; 1 non-default; 6 marginal	
			<i>Target</i>	$I = 27.30$ mM, from sodium acetate 27.30 sodium acetate as 27.30 mM NaCl	
BMRB 6577; PDB 1Z9B	BMRB/PDB: IF2-C1 domain of <i>Bacillus stearothermophilus</i> IF2 [U- ¹³ C; U- ¹⁵ N]; pH 6.4; 307.4 K; 95% H ₂ O/5% D ₂ O; 20 mM KPO ₄ , 200 mM KCl, 5% <i>d</i> ₈ -glycerol; PDB-stated $I = 200$ mM KCl.	HIS75→HIE; HIS121→HIE; <i>Termini</i> : NASN1/CMET135; <i>OXT</i> : present.	<i>Proxy</i>	genion -conc 0.0273	Solute charge -4 ; ions 24 Na ⁺ , 20 Cl ⁻ ; water 39 498; net charge 0.
			<i>Achieved</i>	$pH_{\text{eff}} = 4.50$	
			<i>Final I</i>	22 titratable; 5 non-default; 5 marginal	
			<i>R</i>	$I = 135.00$ mM, from CaCl ₂ 15.00, NaCl 100.00, and ammonium acetate 20.00	
			<i>HH</i>	ammonium acetate as 20.00 mM NaCl	
BMRB 6611; PDB 1ZIT	BMRB: receiver domain of NtrC4 (<i>Aquifex aeolicus</i>), residues 1–121, 1 mM; pH 7.0; 323 K; no D ₂ O fraction listed; 10 mM potassium phosphate. PDB: $I = 40$ mM.	HIS27→HIE; HIS82→HIE; HIS115→HIE; <i>Termini</i> : NMET1/CSER121; <i>OXT</i> : present.	<i>Achieved</i>	$I_{\text{bulk}} = 136.62$ mM (genion : 2 CA; 43 NA, 52 CL)	Solute charge $+5$; ions 2 Ca ²⁺ , 43 Na ⁺ , 52 Cl ⁻ ; water 18 942; net charge 0.
			<i>Final I</i>	$I = 143.59$ mM after neutralisation	
			<i>R</i>	$pH_{\text{eff}} = 6.40$	
			<i>HH</i>	43 titratable; 0 non-default; 0 marginal	
			<i>Target</i>	$I = 225.36$ mM, from KCl 200.00 and phosphate 25.36	
BMRB 6649	BMRB: sporulation-specific N-acetylmuramoyl-L-alanine amidase domain, 1.7 mM [U- ¹⁵ N; U- ¹³ C]; pH 6.9; 303 K; 100% D ₂ O; 50 mM potassium phosphate; $I = 70$ mM.	<i>Termini</i> : NLEU1/CSER79; <i>OXT</i> : present.	<i>Proxy</i>	phosphate as 25.36 mM NaCl	Solute charge -5 ; ions 176 K ⁺ , 22 Na ⁺ , 193 Cl ⁻ ; water 44 792; net charge 0.
			<i>Achieved</i>	$I_{\text{bulk}} = 225.40$ mM (genion : 22 NA; 176 K, 193 CL)	
			<i>Final I</i>	$I = 228.32$ mM after neutralisation	
			<i>R</i>	$pH_{\text{eff}} = 7.00$	
			<i>HH</i>	44 titratable; 0 non-default; 0 marginal	
BMRB 6718; PDB 1ZV6	BMRB/PDB: human dematin headpiece S74E mutant, 1 mM [U- ¹³ C; U- ¹⁵ N]; pH 6.0; 293 K; 90% H ₂ O/10% D ₂ O; 10 mM phosphate; $I = 10$ mM.	HIS33→HIP; <i>Termini</i> : NPRO1/CPHE68; <i>OXT</i> : present.	<i>Target</i>	$I = 17.63$ mM, from phosphate 17.63 phosphate as 17.63 mM NaCl	Solute charge -9 ; ions 15 Na ⁺ , 6 Cl ⁻ ; water 16 914; net charge 0.
			<i>Proxy</i>	phosphate as 17.63 mM NaCl	
			<i>Achieved</i>	$I_{\text{bulk}} = 18.60$ mM (genion : 15 NA, 6 CL)	
			<i>Final I</i>	$I = 32.56$ mM after neutralisation	
			<i>R</i>	$pH_{\text{eff}} = 6.90$	
BMRB 6649	BMRB: sporulation-specific N-acetylmuramoyl-L-alanine amidase domain, 1.7 mM [U- ¹⁵ N; U- ¹³ C]; pH 6.9; 303 K; 100% D ₂ O; 50 mM potassium phosphate; $I = 70$ mM.	<i>Termini</i> : NLEU1/CSER79; <i>OXT</i> : present.	<i>HH</i>	22 titratable; 0 non-default; 0 marginal	Solute charge $+3$; ions 42 Na ⁺ , 45 Cl ⁻ ; water 26 093; net charge 0.
			<i>Target</i>	$I = 82.88$ mM, from phosphate 82.88	
			<i>Proxy</i>	potassium phosphate as 82.88 mM NaCl	
			<i>Achieved</i>	$I_{\text{bulk}} = 82.79$ mM (genion : 42 NA, 45 CL)	
			<i>Final I</i>	$I = 85.75$ mM after neutralisation	
BMRB 6718; PDB 1ZV6	BMRB/PDB: human dematin headpiece S74E mutant, 1 mM [U- ¹³ C; U- ¹⁵ N]; pH 6.0; 293 K; 90% H ₂ O/10% D ₂ O; 10 mM phosphate; $I = 10$ mM.	HIS33→HIP; <i>Termini</i> : NPRO1/CPHE68; <i>OXT</i> : present.	<i>R</i>	$pH_{\text{eff}} = 6.00$	Solute charge $+5$; ions 4 Na ⁺ , 9 Cl ⁻ ; water 17 421; net charge 0.
			<i>HH</i>	26 titratable; 1 non-default; 0 marginal	
			<i>Target</i>	$I = 11.16$ mM, from phosphate 11.16 phosphate as 11.16 mM NaCl	
			<i>Proxy</i>	phosphate as 11.16 mM NaCl	
			<i>Achieved</i>	$I_{\text{bulk}} = 12.23$ mM (genion : 4 NA, 9 CL)	
			<i>Final I</i>	$I = 19.87$ mM after neutralisation	

Record	Provided conditions	Protein preparation	R calculations		GROMACS
BMRB 6763; PDB 2HST	BMRB: middle domain of human eRF1, 1.2 mM; pH 7.0; 298 K; no solution components listed. PDB: 20 mM KPO ₄ , 50 mM KCl, 95% H ₂ O/5% D ₂ O; stated $I = 50$ mM KCl.	HIS31→HIE; HIS41→HIE; HIS60→HIE; <i>Termini</i> : NLEU1/CLEU136; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 7.00$ 40 titratable; 0 non-default; 0 marginal $I = 85.26$ mM, from KCl 50.00 and phosphate 35.26 phosphate as 35.26 mM NaCl $I_{\text{bulk}} = 85.29$ mM (genion : 24 NA; 33 K, 59 CL) $I = 86.78$ mM after neutralisation	Solute charge +2; ions 33 K ⁺ , 24 Na ⁺ , 59 Cl ⁻ ; water 35 562; net charge 0.
BMRB 6781; PDB 2AQE, 2AQF	BMRB: ADA2 SWIRM domain, 0.5 mM [U- ¹³ C; U- ¹⁵ N]; pH 6.5; 278 K; no D ₂ O fraction listed; 50 mM sodium phosphate, 100 mM NaCl, 2 mM DTT, 0.02 mM EDTA. PDB: different D-Tris-HCl sample, pH 7.0; 296 K; 90% H ₂ O/10% D ₂ O; $I = 120$ mM.	HIS53→HIE; <i>Retained</i> : CYS31 and CYS52 free thiols; <i>Termini</i> : NGLY1/CALA90; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Achieved</i>	$pH_{\text{eff}} = 6.50$ 33 titratable; 0 non-default; 1 marginal $I = 166.42$ mM, from NaCl 100.00, phosphate 66.31, and EDTA 0.10 genion -conc 0.16642	Solute charge +7; ions 70 Na ⁺ , 77 Cl ⁻ ; water 22 167; net charge 0.
BMRB 6804; PDB 2B88, 2K1W	BMRB: Ztaq affibody, 1.2 mM [U- ¹³ C; U- ¹⁵ N]; pH* 6.4; 298 K; 90% H ₂ O/10% D ₂ O; 20 mM potassium phosphate, 50 mM NaCl, 0.01% NaN ₃ . PDB: no numeric ionic strength listed.	<i>Termini</i> : NVAL1/CLYS58; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.44$ 16 titratable; 0 non-default; 0 marginal $I = 75.36$ mM, from NaCl 50.00 and phosphate 25.36 phosphate as 25.36 mM NaCl $I_{\text{bulk}} = 76.39$ mM (genion : 25 NA, 24 CL) $I = 77.98$ mM after neutralisation	Solute charge -1; ions 25 Na ⁺ , 24 Cl ⁻ ; water 16 681; net charge 0.
BMRB 6805; PDB 2B89	BMRB: anti-Ztaq affibody, 1.5 mM [U- ¹³ C; U- ¹⁵ N]; pH* 6.4; 298 K; 90% H ₂ O/10% D ₂ O; 20 mM potassium phosphate, 50 mM NaCl, 0.01% NaN ₃ . PDB: no numeric ionic strength listed.	<i>Termini</i> : NVAL1/CLYS58; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.44$ 16 titratable; 0 non-default; 0 marginal $I = 75.36$ mM, from NaCl 50.00 and phosphate 25.36 phosphate as 25.36 mM NaCl $I_{\text{bulk}} = 75.99$ mM (genion : 32 NA, 33 CL) $I = 77.18$ mM after neutralisation	Solute charge +1; ions 32 Na ⁺ , 33 Cl ⁻ ; water 22 506; net charge 0.
BMRB 6816; PDB 2AHQ	BMRB: RpoN; 0.8 mM [U- ¹⁵ N]; pH 6.8; 298 K; 90% H ₂ O/10% D ₂ O; 50 mM HEPES, 250 mM NaCl, 1 mM EDTA; stated $I = 300$ mM.	<i>Termini</i> : NTHR1/CILE76; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.80$ 33 titratable; 0 non-default; 0 marginal $I = 263.02$ mM, from NaCl 250.00, HEPES 7.55, and EDTA 5.47 HEPES and EDTA as 13.02 mM NaCl $I_{\text{bulk}} = 263.02$ mM (genion : 135 NA, 140 CL) $I = 267.89$ mM after neutralisation	Solute charge +5; ions 135 Na ⁺ , 140 Cl ⁻ ; water 25 925; net charge 0.
BMRB 6854; PDB 2B86	BMRB: Nck SH3 domain 1; 0.6 mM [U- ¹³ C; U- ¹⁵ N]; pH 6.5; 298 K; 50 mM sodium phosphate, 50 mM EDTA. PDB: 95% H ₂ O/5% D ₂ O; stated $I = 50$ mM phosphate.	<i>Termini</i> : NMET1/CLYS59; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.50$ 24 titratable; 0 non-default; 0 marginal $I = 321.47$ mM, from phosphate 66.31 and EDTA 255.16 phosphate and EDTA as 321.47 mM NaCl $I_{\text{bulk}} = 322.75$ mM (genion : 101 NA, 101 CL) $I = 322.75$ mM after neutralisation	Solute charge 0; ions 101 Na ⁺ , 101 Cl ⁻ ; water 16 438; net charge 0.
BMRB 6865; PDB 2BBX	BMRB: TRAP-TSR; 0.5 mM [U- ¹³ C; U- ¹⁵ N]; pH 6.6; 283 K; no D ₂ O fraction listed; 20 mM Bis-Tris.	HIS31→HIE; <i>Bonded</i> : CYS5–CYS34, CYS14–CYS42, and CYS18–CYS47 S–S; residues CYX; <i>Termini</i> : NGLY1/CPRO49; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.60$ 23 titratable; 6 non-default; 0 marginal $I = 8.40$ mM, from Bis-Tris Bis-Tris as 8.40 mM NaCl $I_{\text{bulk}} = 9.01$ mM (genion : 8 NA, 5 CL) $I = 11.71$ mM after neutralisation	Solute charge -3; ions 8 Na ⁺ , 5 Cl ⁻ ; water 29 801; net charge 0.
BMRB 6912; PDB 2I9H	BMRB: reduced yeast thioredoxin 1; 1 mM [U- ¹³ C; U- ¹⁵ N]; pH 7.0; 308 K; 90% H ₂ O/10% D ₂ O; 3 mM 2-mercaptoethanol. PDB: 298 K; 10 mM DTT; 20 mM phosphate; stated $I = 20$ mM.	<i>Retained</i> : CYS30 and CYS33 free thiols; <i>Termini</i> : NMET1/CALA103; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 7.00$ 29 titratable; 2 non-default; 1 marginal $I = 35.26$ mM, from phosphate phosphate as 35.26 mM NaCl $I_{\text{bulk}} = 34.31$ mM (genion : 13 NA, 10 CL) $I = 39.45$ mM after neutralisation	Solute charge -3; ions 13 Na ⁺ , 10 Cl ⁻ ; water 15 346; net charge 0.

Record	Provided conditions	Protein preparation	R calculations		GROMACS
BMRB 6932; PDB 2FM4	BMRB: cent-I; 1.0 mM [U- ¹³ C; U- ¹⁵ N]; pH 7.0; 298 K; 50 mM KCl, 20 mM potassium phosphate; stated $I = 0.07$ M. PDB: 95% H ₂ O/10% D ₂ O.	HIS35→HIE; HIS57→HIE; HIS72→HIE; HIS105→HIE; <i>Retained</i> : CYS83, CYS84, and CYS88 free thiols; <i>Termini</i> : NALA1/CGLU128; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 7.00$ 42 titratable; 0 non-default; 1 marginal $I = 85.26$ mM, from KCl 50.00 and phosphate 35.26 potassium phosphate as 35.26 mM NaCl $I_{\text{bulk}} = 85.60$ mM (genion : 12 NA; 23 K, 28 CL) $I = 96.30$ mM after neutralisation	Solute charge −7; ions 23 K ⁺ , 12 Na ⁺ , 28 Cl [−] ; water 17 151; net charge 0.
BMRB 6964; PDB 2FVT	BMRB: Rpa2829; 1.2 mM [U- ¹³ C; U- ¹⁵ N]; pH 5.0; 293 K; 20 mM ammonium acetate, 100 mM NaCl, 5 mM CaCl ₂ . PDB: 95% H ₂ O/10% D ₂ O; 10 mM DTT.	HIS9→HIP; HIS31→HIP; HIS130→HIP; HIS131→HIP; HIS132→HIP; HIS133→HIP; HIS134→HIP; <i>Termini</i> : NMET1/CHIP135; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 5.00$ 39 titratable; 8 non-default; 0 marginal $I = 127.69$ mM, from CaCl ₂ 15.00, NaCl 100.00, and acetate 12.69 ammonium acetate as 12.69 mM NaCl $I_{\text{bulk}} = 129.15$ mM (genion : 3 CA; 60 NA, 74 CL) $I = 136.64$ mM after neutralisation	Solute charge +8; ions 3 Ca ²⁺ , 60 Na ⁺ , 74 Cl [−] ; water 28 647; net charge 0.
BMRB 6976; PDB 2FJ6	BMRB: hypothetical UPF0346 protein yozE; 1.5 mM [U- ¹³ C; U- ¹⁵ N]; pH 5.5; 293 K; 95% H ₂ O/5% D ₂ O; 100 mM NaCl, 5 mM CaCl ₂ , 20 mM ammonium acetate, 10 mM DTT, 0.02% NaN ₃ ; stated $I = 100$ mM.	HIS6→HIP; HIS13→HIP; HIS31→HIP; HIS41→HIP; HIS55→HIP; HIS72→HIP; HIS77→HIP; HIS78→HIP; HIS79→HIP; HIS80→HIP; HIS81→HIP; <i>Termini</i> : NMET1/CHIP82; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 5.50$ 42 titratable; 12 non-default; 0 marginal $I = 131.92$ mM, from CaCl ₂ 15.00, NaCl 100.00, and acetate 16.92 ammonium acetate as 16.92 mM NaCl $I_{\text{bulk}} = 133.89$ mM (genion : 2 CA; 38 NA, 47 CL) $I = 143.12$ mM after neutralisation	Solute charge +5; ions 2 Ca ²⁺ , 38 Na ⁺ , 47 Cl [−] ; water 17 141; net charge 0.
BMRB 6981; PDB 2G0U	BMRB: BsaL; 0.8 mM [U- ¹⁵ N; U- ¹³ C]; pH 7.0; no D ₂ O fraction listed; no solution components listed. PDB: 298 K; stated $I = 20$ mM.	HIS87→HIP; HIS88→HIE; HIS89→HIE; HIS90→HIE; HIS91→HIE; <i>Termini</i> : NMET1/CHIE92; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 7.00$ 30 titratable; 1 non-default; 1 marginal $I = 20.00$ mM, from deposited scalar scalar as 20.00 mM NaCl $I_{\text{bulk}} = 19.63$ mM (genion : 25 NA, 22 CL) $I = 20.97$ mM after neutralisation	Solute charge −3; ions 25 Na ⁺ , 22 Cl [−] ; water 60 497; net charge 0.
BMRB 6989; PDB 2G1D	BMRB: ribosomal protein S24E; 1.5 mM [U- ¹³ C; U- ¹⁵ N]; pH 6.5; 298 K; 90% H ₂ O/10% D ₂ O; 50 mM potassium phosphate, 0.1 mM NaN ₃ .	HIS43→HIP; HIS63→HIE; <i>Termini</i> : NMET1/CLYS98; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.50$ 48 titratable; 1 non-default; 1 marginal $I = 66.31$ mM, from phosphate potassium phosphate as 66.31 mM NaCl $I_{\text{bulk}} = 65.77$ mM (genion : 27 NA, 30 CL) $I = 69.43$ mM after neutralisation	Solute charge +3; ions 27 Na ⁺ , 30 Cl [−] ; water 21 565; net charge 0.
BMRB 7000; PDB 2A7Y	BMRB: Rv2302; 2 mM [U- ¹³ C; U- ¹⁵ N]; pH 7.1; 298 K; 90% H ₂ O/10% D ₂ O; 100 mM KCl, 20 mM potassium phosphate, 2 mM DTT.	HIS3→HIE; HIS5→HIE; HIS22→HIE; HIS25→HIE; HIS49→HIE; HIS67→HIE; HIS80→HIE; <i>Termini</i> : NGLY1/CTHR83; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 7.10$ 30 titratable; 0 non-default; 0 marginal $I = 137.48$ mM, from KCl 100.00 and phosphate 37.48 potassium phosphate as 37.48 mM NaCl $I_{\text{bulk}} = 138.82$ mM (genion : 13 NA; 38 K, 48 CL) $I = 143.16$ mM after neutralisation	Solute charge −3; ions 38 K ⁺ , 13 Na ⁺ , 48 Cl [−] ; water 18 279; net charge 0.
BMRB 7057; PDB 1ZWV, 2COO	BMRB: hbSBD [U- ¹³ C; U- ¹⁵ N], 1–2 mM; pH 7.5; 295 K; no D ₂ O fraction listed; 50 mM phosphate buffer, 100 mM NaCl; $I = 100$ mM. PDB: 50 mM phosphate buffer; $I = 100$ mM NaCl.	<i>Termini</i> : NGLY1/CGLU52; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 7.50$ 20 titratable; 0 non-default; 0 marginal $I = 216.10$ mM, from phosphate 116.10 and NaCl 100.00 phosphate buffer as 116.10 mM NaCl $I_{\text{bulk}} = 216.81$ mM (genion : 91 NA, 93 CL) $I = 219.20$ mM after neutralisation	Solute charge +2; ions 91 Na ⁺ , 93 Cl [−] ; water 22 392; net charge 0.
BMRB 7094; PDB 2GV1	BMRB: Escherichia coli acylphosphatase, 0.5 mM; pH 4.95; 310 K. PDB: 90% H ₂ O/10% D ₂ O; 50 mM sodium phosphate, 30 mM deuterated sodium acetate, 50 mM NaCl; $I = 65$ mM.	HIS80→HIP; HIS81→HIP; <i>Bonded</i> : CYS5–CYS49 S–S; residues CYX; <i>Termini</i> : NMET1/CARG92; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 4.95$ 33 titratable; 4 non-default; 3 marginal $I = 118.70$ mM, from phosphate 50.47, acetate 18.23, and NaCl 50.00 sodium phosphate and deuterated sodium acetate as 68.70 mM NaCl $I_{\text{bulk}} = 118.39$ mM (genion : 41 NA, 45 CL) $I = 124.16$ mM after neutralisation	Solute charge +4; ions 41 Na ⁺ , 45 Cl [−] ; water 18 246; net charge 0.

Record	Provided conditions	Protein preparation	R calculations		GROMACS
BMRB 7104; PDB 1LWR	BMRB: F3 module 2 of NCAM [U- ¹⁵ N], 1 mM; pH 7.27; 298 K; no D ₂ O fraction listed; 10 mM sodium phosphate, 30 mM NaCl; <i>I</i> = 30 mM. PDB: 10 mM sodium phosphate, 30 mM NaCl; <i>I</i> = 40 mM.	HIS34→HIE; HIS58→HID; HIS87→HIE; <i>Termini</i> : NGLY1/CGLN95; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 7.27$ 32 titratable; 0 non-default; 0 marginal <i>I</i> = 50.69 mM, from NaCl 30.00 and phosphate 20.69 sodium phosphate as 20.69 mM NaCl <i>I</i> _{bulk} = 50.52 mM (genion : 22 NA, 22 CL) <i>I</i> = 50.52 mM after neutralisation	Solute charge 0; ions 22 Na ⁺ , 22 Cl ⁻ ; water 23 120; net charge 0.
BMRB 7109; PDB 2GZZ	BMRB: TrxA [U-95% ¹³ C; U-95% ¹⁵ N], 1.00 mM; pH 6.85; 298 K; no D ₂ O fraction listed; <i>I</i> = 0.06 M. PDB: pH 6.8; 20 mM Tris-HCl buffer.	HIS103→HIE; <i>Bonded</i> : CYS29–CYS32 S–S; residues CYX; <i>Termini</i> : NMET1/CLEU104; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.85$ 33 titratable; 2 non-default; 0 marginal <i>I</i> = 18.84 mM, from Tris 18.84 Tris-HCl as 18.84 mM NaCl <i>I</i> _{bulk} = 20.07 mM (genion : 14 NA, 6 CL) <i>I</i> = 33.45 mM after neutralisation	Solute charge −8; ions 14 Na ⁺ , 6 Cl ⁻ ; water 15 752; net charge 0.
BMRB 7118; PDB 2GLE	BMRB: Neurabin SAM domain [U-95% ¹³ C; U-99% ¹⁵ N] and [U-99% ¹⁵ N] samples; 1.5 mM; pH 6.8; 298 K; selected condition 90% H ₂ O/10% D ₂ O; 20 mM NaPi, 50 mM NaCl. PDB: <i>I</i> = 50 mM NaCl.	HIS2→HIE; HIS5→HIE; HIS14→HIE; <i>Termini</i> : NGLY1/CALA74; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.80$ 22 titratable; 0 non-default; 1 marginal <i>I</i> = 81.20 mM, from phosphate 31.20 and NaCl 50.00 NaPi as 31.20 mM NaCl <i>I</i> _{bulk} = 81.62 mM (genion : 23 NA, 24 CL) <i>I</i> = 83.39 mM after neutralisation	Solute charge +1; ions 23 Na ⁺ , 24 Cl ⁻ ; water 15 075; net charge 0.
BMRB 7127; PDB 2H41, 2H45	BMRB: 2F3 [U- ¹⁵ N] and [U- ¹³ C; U- ¹⁵ N] samples; 1.5–2.5 mM; pH 7.0; 303 K; 95% H ₂ O/5% D ₂ O; 20 mM NaCl, 20 mM NaPi; <i>I</i> = 160 mM. PDB: 20 mM NaCl, 20 mM Na ₂ HPO ₄ .	HIS6→HIE; <i>Termini</i> : NGLY1/CTHR95; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 7.00$ 23 titratable; 0 non-default; 0 marginal <i>I</i> = 55.26 mM, from NaCl 20.00 and phosphate 35.26 NaPi as 35.26 mM NaCl <i>I</i> _{bulk} = 54.02 mM (genion : 30 NA, 19 CL) <i>I</i> = 69.66 mM after neutralisation	Solute charge −11; ions 30 Na ⁺ , 19 Cl ⁻ ; water 18 547; net charge 0.
BMRB 7175; PDB 2HEQ	BMRB: YorP [U-100% ¹³ C; U-100% ¹⁵ N], 1.0 mM; pH 5.5; 293 K; 20 mM ammonium acetate, 100 mM NaCl; <i>I</i> = 100 mM. PDB: 95% H ₂ O/5% D ₂ O; 5 mM CaCl ₂ , 10 mM DTT; 298 K.	HIS30→HIP; HIS31→HIP; HIS41→HIP; HIS43→HIP; HIS64→HIP; HIS79→HIP; HIS80→HIP; HIS81→HIP; HIS82→HIP; HIS83→HIP; <i>Retained</i> : CYS28 free thiol; <i>Termini</i> : NMET1/CHIP84; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 5.50$ 37 titratable; 11 non-default; 0 marginal <i>I</i> = 135.00 mM, from CaCl ₂ 15.00, ammonium acetate 20.00, and NaCl 100.00 ammonium acetate as 20.00 mM NaCl <i>I</i> _{bulk} = 133.76 mM (genion : 3 CA; 83 NA, 97 CL) <i>I</i> = 139.58 mM after neutralisation	Solute charge +8; ions 3 Ca ²⁺ , 83 Na ⁺ , 97 Cl ⁻ ; water 36 765; net charge 0.
BMRB 7181; PDB 2H3J	BMRB: Hypothetical protein PA4359 [U- ¹³ C; U- ¹⁵ N], 0.7 mM; pH 7.0; 298 K; 90% H ₂ O/10% D ₂ O; 10 mM Tris, 300 mM NaCl, 5% glycerol, 0.01% NaN ₃ , 1 mM benzamidine. PDB: <i>I</i> not listed.	<i>Termini</i> : NMET1/CASP75; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 7.00$ 18 titratable; 0 non-default; 0 marginal <i>I</i> = 309.20 mM, from Tris 9.20 and NaCl 300.00 Tris as 9.20 mM NaCl <i>I</i> _{bulk} = 309.58 mM (genion : 87 NA, 93 CL) <i>I</i> = 320.26 mM after neutralisation	Solute charge +6; ions 87 Na ⁺ , 93 Cl ⁻ ; water 14 901; net charge 0.
BMRB 7185; PDB 2GUT	BMRB: Positive cofactor 2 glutamine/Q-rich-associated protein [U- ¹³ C; U- ¹⁵ N], 1.0 mM; pH 6.8; 298 K; 90% H ₂ O/10% D ₂ O; 30 mM NaCl, 10 mM phosphate buffer; <i>I</i> = 30 mM. PDB: <i>I</i> = 30 mM NaCl.	HIS33→HIE; HIS43→HIE; HIS64→HIE; HIS69→HIE; <i>Termini</i> : NGLY1/CVAL77; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.80$ 29 titratable; 0 non-default; 2 marginal <i>I</i> = 45.60 mM, from phosphate 15.60 and NaCl 30.00 phosphate buffer as 15.60 mM NaCl <i>I</i> _{bulk} = 44.81 mM (genion : 24 NA, 29 CL) <i>I</i> = 49.48 mM after neutralisation	Solute charge +5; ions 24 Na ⁺ , 29 Cl ⁻ ; water 29 010; net charge 0.
BMRB 7223; PDB 2HJ8	BMRB: Interferon-induced 17 kDa protein [U- ¹³ C; U- ¹⁵ N] and [U-5% ¹³ C; U- ¹⁵ N] samples; 0.81–0.9 mM; pH 6.5; 293 K; 95% H ₂ O/5% D ₂ O; 50 mM ammonium citrate, 5 mM CaCl ₂ , 0.02% NaN ₃ ; <i>I</i> = 50 mM. PDB: 50 mM ammonium citrate, 5 mM CaCl ₂ .	HIS29→HIE; HIS83→HIE; HIS84→HIE; HIS85→HIE; HIS86→HIE; HIS87→HIE; <i>Termini</i> : NMET1/CHIE88; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.50$ 30 titratable; 0 non-default; 5 marginal <i>I</i> = 247.13 mM, from citrate 232.13 and CaCl ₂ 15.00 ammonium citrate as 232.13 mM NaCl <i>I</i> _{bulk} = 246.53 mM (genion : 2 CA; 107 NA, 109 CL) <i>I</i> = 249.87 mM after neutralisation	Solute charge −2; ions 2 Ca ²⁺ , 107 Na ⁺ , 109 Cl ⁻ ; water 23 631; net charge 0.

Record	Provided conditions	Protein preparation	R calculations		GROMACS
BMRB 7350; PDB 2BZB	BMRB: conserved domain protein; pH 7.0; 298 K; 90% H ₂ O/10% D ₂ O; <i>I</i> = 150 mM; no salt or buffer components listed.	HIS24→HIE; HIS28→HIE; HIS53→HIE; <i>Termini</i> : NMET1/CLYS54; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 7.00$ 25 titratable; 0 non-default; 0 marginal <i>I</i> = 150.00 mM, from a deposited scalar value deposited ionic-strength scalar as 150.00 mM NaCl <i>I</i> _{bulk} = 149.72 mM (genion : 56 NA, 61 CL) <i>I</i> = 156.40 mM after neutralisation	Solute charge +5; ions 56 Na ⁺ , 61 Cl ⁻ ; water 20 079; net charge 0.
BMRB 7362; PDB 2NWT	BMRB/PDB: UPF0165 protein AF_2212; 1.2 mM [U- ¹³ C; U- ¹⁵ N]; pH 5.5; 298 K; 95% H ₂ O/5% D ₂ O; 100 mM NaCl, 5 mM CaCl ₂ , 20 mM ammonium acetate, 10 mM DTT, 0.02% NaN ₃ ; stated <i>I</i> = 100 mM.	HIS64→HIP; HIS65→HIP; HIS66→HIP; HIS67→HIP; HIS68→HIP; <i>Termini</i> : NMET1/CHIP69; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 5.50$ 32 titratable; 6 non-default; 0 marginal <i>I</i> = 135.00 mM, from NaCl 100.00, CaCl ₂ 15.00, and ammonium acetate 20.00 ammonium acetate as 20.00 mM NaCl <i>I</i> _{bulk} = 134.38 mM (genion : 5 CA; 132 NA, 146 CL) <i>I</i> = 135.74 mM after neutralisation	Solute charge +4; ions 5 Ca ²⁺ , 132 Na ⁺ , 146 Cl ⁻ ; water 59 066; net charge 0.
BMRB 7366; PDB 2NOC	BMRB: putative periplasmic protein; 1.1 mM; pH 4.5; 298 K; 90% H ₂ O/10% D ₂ O; 10 mM Tris, 100 mM NaCl, 0.02% NaN ₃ . PDB: no numeric ionic strength listed.	GLU23→GLH; GLU36→GLH; GLU48→GLH; GLU64→GLH; GLU78→GLH; HIS82→HIP; HIS94→HIP; HIS95→HIP; HIS96→HIP; HIS97→HIP; HIS98→HIP; <i>Termini</i> : NMET1/CHIP99; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 4.50$ 40 titratable; 12 non-default; 6 marginal <i>I</i> = 110.00 mM, from NaCl 100.00 and Tris 10.00 Tris as 10.00 mM NaCl <i>I</i> _{bulk} = 109.82 mM (genion : 64 NA, 76 CL) <i>I</i> = 120.11 mM after neutralisation	Solute charge +12; ions 64 Na ⁺ , 76 Cl ⁻ ; water 31 135; net charge 0.
BMRB 7371; PDB 2OA4	BMRB/PDB: SiR5, 1.0–1.15 mM labelled protein; pH 6.5; 293 K; 95% H ₂ O/5% D ₂ O; <i>I</i> = 0.1 M NaCl. BMRB sample 1 lists NaCl 0.1 mM.	HIS77→HIP; HIS96→HIP; HIS97→HIE; HIS98→HIP; HIS99→HIE; HIS100→HIE; <i>Termini</i> : NMET1/CHIP101; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Achieved</i>	$pH_{\text{eff}} = 6.50$ 36 titratable; 4 non-default; 2 marginal <i>I</i> = 100.00 mM, from a deposited scalar value genion -conc 0.1 $pH_{\text{eff}} = 6.00$	Solute charge +9; ions 45 Na ⁺ , 54 Cl ⁻ ; water 23 856; net charge 0.
BMRB 10005; PDB 2D49	BMRB: ChitinaseC chitin-binding domain [U- ¹³ C; U- ¹⁵ N], 1.0 mM; pH 6.0; 303 K. PDB: selected 90% H ₂ O/10% D ₂ O; 20 mM NAPI/phosphate, 1 mM DTT, 0.02% NaN ₃ .	<i>Bonded</i> : CYS4–CYS50 S–S; residues CYX; <i>Termini</i> : NMET1/CSER54; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	13 titratable; 2 non-default; 0 marginal <i>I</i> = 22.32 mM, from phosphate 22.32 phosphate as 22.32 mM NaCl <i>I</i> _{bulk} = 21.95 mM (genion : 6 NA, 7 CL) <i>I</i> = 23.78 mM after neutralisation	Solute charge +1; ions 6 Na ⁺ , 7 Cl ⁻ ; water 14 508; net charge 0.
BMRB 10028; PDB 2DWV	BMRB/PDB: WW domain [U- ¹³ C; U- ¹⁵ N], 1 mM; pH 7.0; 298 K; 90% H ₂ O/10% D ₂ O; 20 mM d-Tris HCl, 100 mM NaCl, 1 mM d-DTT, 0.02% NaN ₃ ; <i>I</i> = 0.12 M.	HIS33→HIE; HIS42→HIE; <i>Termini</i> : NGLY1/CGLY49; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 7.00$ 17 titratable; 0 non-default; 0 marginal <i>I</i> = 118.40 mM, from NaCl 100.00 and Tris 18.40 Tris as 18.40 mM NaCl <i>I</i> _{bulk} = 118.21 mM (genion : 56 NA, 55 CL) <i>I</i> = 119.28 mM after neutralisation	Solute charge −1; ions 56 Na ⁺ , 55 Cl ⁻ ; water 24 707; net charge 0.
BMRB 10047; PDB 2ECC	BMRB/PDB: homeobox domain [U- ¹³ C; U- ¹⁵ N], 1.14 mM; pH 7.0; 298 K; 90% H ₂ O/10% D ₂ O; 20 mM d-Tris-HCl, 100 mM NaCl, 1 mM d-DTT, 0.02% NaN ₃ ; <i>I</i> = 120 mM.	HIS60→HIE; <i>Retained</i> : CYS25 free thiol; <i>Termini</i> : NGLY1/CGLY76; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 7.00$ 25 titratable; 0 non-default; 0 marginal <i>I</i> = 118.40 mM, from NaCl 100.00 and Tris 18.40 Tris as 18.40 mM NaCl <i>I</i> _{bulk} = 119.28 mM (genion : 62 NA, 68 CL) <i>I</i> = 125.05 mM after neutralisation	Solute charge +6; ions 62 Na ⁺ , 68 Cl ⁻ ; water 27 101; net charge 0.
BMRB 10050; PDB 1WJU	BMRB/PDB: ubiquitin-like domain [U- ¹³ C; U- ¹⁵ N], 1 mM; pH 7.0; 298 K; 90% H ₂ O/10% D ₂ O; 20 mM d-Tris-HCl, 100 mM NaCl, 1 mM d-DTT, 0.02% NaN ₃ ; <i>I</i> = 120 mM.	HIS39→HIE; HIS82→HIE; <i>Termini</i> : NGLY1/CGLY100; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 7.00$ 31 titratable; 0 non-default; 0 marginal <i>I</i> = 118.40 mM, from NaCl 100.00 and Tris 18.40 Tris as 18.40 mM NaCl <i>I</i> _{bulk} = 118.39 mM (genion : 101 NA, 107 CL) <i>I</i> = 121.90 mM after neutralisation	Solute charge +6; ions 101 Na ⁺ , 107 Cl ⁻ ; water 45 145; net charge 0.

Record	Provided conditions	Protein preparation	R calculations		GROMACS
BMRB 10056; PDB 1WFY	BMRB/PDB: RBD [U- ¹³ C; U- ¹⁵ N], 0.18 mM; pH 7.5; 298 K; 90% H ₂ O/10% D ₂ O; 20 mM d-Tris-HCl, 100 mM NaCl, 1 mM d-DTT, 0.02% NaN ₃ ; <i>I</i> = 120 mM.	HIS51→HIE; HIS61→HIE; <i>Termini</i> : NGLY1/CGLY104; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i>	$pH_{\text{eff}} = 7.50$ 28 titratable; 0 non-default; 0 marginal <i>I</i> = 115.68 mM, from NaCl 100.00 and Tris 15.68 <i>Proxy</i> <i>Achieved</i> Tris as 15.68 mM NaCl <i>I</i> _{bulk} = 115.69 mM (genion : 105 NA, 104 CL)	Solute charge −1; ions 105 Na ⁺ , 104 Cl [−] ; water 48 684; net charge 0.
BMRB 10061; PDB 1WJL	BMRB/PDB: PDZ domain [U- ¹³ C; U- ¹⁵ N], 1.1 mM; pH 6.0; 298 K; 90% H ₂ O/10% D ₂ O; 20 mM phosphate, 100 mM NaCl, 1 mM d-DTT, 0.02% NaN ₃ ; <i>I</i> = 120 mM.	HIS69→HIP; <i>Termini</i> : NGLY1/CSER90; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i>	$pH_{\text{eff}} = 6.00$ 16 titratable; 1 non-default; 0 marginal <i>I</i> = 122.32 mM, from NaCl 100.00 and phosphate 22.32 <i>Proxy</i> <i>Achieved</i> phosphate as 22.32 mM NaCl <i>I</i> _{bulk} = 121.68 mM (genion : 64 NA, 67 CL)	Solute charge +3; ions 64 Na ⁺ , 67 Cl [−] ; water 28 245; net charge 0.
BMRB 10063; PDB 1UE9	BMRB/PDB: SH3 domain [U- ¹³ C; U- ¹⁵ N], 1.4 mM; pH 6.0; 298 K; 90% H ₂ O/10% D ₂ O; 20 mM phosphate buffer, 100 mM NaCl, 0.02% NaN ₃ ; <i>I</i> = 120 mM.	HIS63→HIP; <i>Termini</i> : NGLY1/CGLY80; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i>	$pH_{\text{eff}} = 6.00$ 16 titratable; 1 non-default; 1 marginal <i>I</i> = 122.32 mM, from NaCl 100.00 and phosphate 22.32 <i>Proxy</i> <i>Achieved</i> phosphate as 22.32 mM NaCl <i>I</i> _{bulk} = 122.92 mM (genion : 44 NA, 50 CL)	Solute charge +6; ions 44 Na ⁺ , 50 Cl [−] ; water 18 888; net charge 0.
BMRB 10069; PDB 1UJS	BMRB/PDB: VHP domain [U- ¹³ C; U- ¹⁵ N], 1.2 mM; pH 6.0; 298 K; 90% H ₂ O/10% D ₂ O; 20 mM phosphate buffer, 100 mM NaCl, 0.02% NaN ₃ ; <i>I</i> = 120 mM.	HIS47→HID; <i>Termini</i> : NGLY1/CGLY88; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i>	$pH_{\text{eff}} = 6.00$ 31 titratable; 0 non-default; 1 marginal <i>I</i> = 122.32 mM, from phosphate 22.32 and NaCl 100.00 <i>Proxy</i> <i>Achieved</i> phosphate as 22.32 mM NaCl <i>I</i> _{bulk} = 121.70 mM (genion : 52 NA, 57 CL)	Solute charge +5; ions 52 Na ⁺ , 57 Cl [−] ; water 22 681; net charge 0.
BMRB 10077; PDB 1WI8	BMRB: RNA recognition motif [U- ¹³ C; U- ¹⁵ N], 1.0 mM; pH 7.0; 298 K; 90% H ₂ O/10% D ₂ O; 20 mM d-Tris-HCl, 100 mM NaCl, 1 mM d-DTT, 0.02% NaN ₃ ; <i>I</i> = 100 mM.	<i>Termini</i> : NGLY1/CGLY104; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i>	$pH_{\text{eff}} = 7.00$ 32 titratable; 0 non-default; 0 marginal <i>I</i> = 118.40 mM, from NaCl 100.00 and Tris 18.40 <i>Proxy</i> <i>Achieved</i> d-Tris-HCl as 18.40 mM NaCl <i>I</i> _{bulk} = 118.75 mM (genion : 95 NA, 91 CL)	Solute charge −4; ions 95 Na ⁺ , 91 Cl [−] ; water 40 910; net charge 0.
BMRB 10087; PDB 1WH3	BMRB: ubiquitin-like domain [U- ¹³ C; U- ¹⁵ N], 1.1 mM; pH 7.5; 296 K; 90% H ₂ O/10% D ₂ O; 20 mM d-Tris HCl, 100 mM NaCl, 1 mM d-DTT, 0.02% NaN ₃ ; <i>I</i> = 120 mM.	<i>Termini</i> : NGLY1/CGLY87; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i>	$pH_{\text{eff}} = 7.50$ 18 titratable; 0 non-default; 0 marginal <i>I</i> = 115.68 mM, from NaCl 100.00 and Tris 15.68 <i>Proxy</i> <i>Achieved</i> d-Tris HCl as 15.68 mM NaCl <i>I</i> _{bulk} = 115.51 mM (genion : 66 NA, 66 CL)	Solute charge 0; ions 66 Na ⁺ , 66 Cl [−] ; water 30 753; net charge 0.
BMRB 10095; PDB 1WH9	BMRB/PDB: KH domain [U- ¹³ C; U- ¹⁵ N], 0.6 mM; pH 7.0; 296 K; 90% H ₂ O/10% D ₂ O; 20 mM d-Tris-HCl, 200 mM NaCl, 1 mM d-DTT, 0.02% NaN ₃ ; <i>I</i> = 220 mM.	<i>Termini</i> : NGLY1/CGLY92; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i>	$pH_{\text{eff}} = 7.00$ 28 titratable; 0 non-default; 0 marginal <i>I</i> = 218.40 mM, from Tris 18.40 and NaCl 200.00 <i>Proxy</i> <i>Achieved</i> d-Tris-HCl as 18.40 mM NaCl <i>I</i> _{bulk} = 218.45 mM (genion : 146 NA, 147 CL)	Solute charge +1; ions 146 Na ⁺ , 147 Cl [−] ; water 35 621; net charge 0.
BMRB 10100; PDB 1UEP	BMRB: PDZ domain [U- ¹³ C; U- ¹⁵ N], 1.3 mM; pH 7.0; 298 K; 90% H ₂ O/10% D ₂ O; 20 mM Tris-HCl, 100 mM NaCl, 1 mM d-DTT, 0.02% NaN ₃ ; <i>I</i> = 120 mM. PDB: pH 7.0; 298 K; no D ₂ O fraction listed.	HIS14→HIE; HIS54→HIE; HIS72→HID; HIS80→HIE; HIS81→HIE; <i>Termini</i> : NGLY1/CGLY103; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i>	$pH_{\text{eff}} = 7.00$ 32 titratable; 0 non-default; 1 marginal <i>I</i> = 118.40 mM, from Tris 18.40 and NaCl 100.00 <i>Proxy</i> <i>Achieved</i> Tris-HCl as 18.40 mM NaCl <i>I</i> _{bulk} = 118.57 mM (genion : 51 NA, 52 CL)	Solute charge +1; ions 51 Na ⁺ , 52 Cl [−] ; water 22 807; net charge 0.
			<i>Final I</i>	<i>I</i> = 119.73 mM after neutralisation	

Record	Provided conditions	Protein preparation	R calculations		GROMACS
BMRB 10111; PDB 1V63	BMRB/PDB: HMG box domain [U- ¹³ C; U- ¹⁵ N], 1.2 mM; pH 6.0; 298 K; 90% H ₂ O/10% D ₂ O; 20 mM phosphate buffer, 100 mM NaCl, 1 mM d-DTT, 0.02% NaN ₃ ; $I = 120$ mM.	HIS31→HIP; HIS55→HIP; HIS70→HIP; <i>Termini</i> : NGLY1/CGLY101; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.00$ 37 titratable; 3 non-default; 3 marginal $I = 122.32$ mM, from NaCl 100.00 and phosphate 22.32 phosphate as 22.32 mM NaCl $I_{\text{bulk}} = 122.97$ mM (genion : 70 NA, 81 CL) $I = 132.63$ mM after neutralisation	Solute charge +11; ions 70 Na ⁺ , 81 Cl ⁻ ; water 30 443; net charge 0.
BMRB 10112; PDB 1V86	BMRB/PDB: ubiquitin domain [U- ¹³ C; U- ¹⁵ N], 0.39 mM; pH 6.0; 298 K; 90% H ₂ O/10% D ₂ O; 20 mM sodium phosphate, 100 mM NaCl, 1 mM d-DTT, 0.02% NaN ₃ ; $I = 120$ mM.	HIS29→HIP; HIS47→HIP; <i>Termini</i> : NGLY1/CGLY95; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.00$ 26 titratable; 2 non-default; 2 marginal $I = 122.32$ mM, from phosphate 22.32 and NaCl 100.00 sodium phosphate as 22.32 mM NaCl $I_{\text{bulk}} = 122.85$ mM (genion : 86 NA, 92 CL) $I = 127.13$ mM after neutralisation	Solute charge +6; ions 86 Na ⁺ , 92 Cl ⁻ ; water 37 490; net charge 0.
BMRB 10118; PDB 2EXD	BMRB/PDB: PH0471 C-terminal domain [U- ¹³ C; U- ¹⁵ N], 0.7 mM; pH 5.0; 298 K; 95% H ₂ O/5% D ₂ O; 25 mM sodium phosphate, 300 mM NaCl, 1 mM EDTA; $I = 300$ mM.	GLU15→GLH; HIS30→HIP; <i>Termini</i> : NARG1/CGLU74; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 5.00$ 33 titratable; 2 non-default; 0 marginal $I = 328.47$ mM, from phosphate 25.27, NaCl 300.00, and EDTA 3.20 sodium phosphate and EDTA as 28.47 mM NaCl $I_{\text{bulk}} = 328.66$ mM (genion : 118 NA, 120 CL) $I = 331.45$ mM after neutralisation	Solute charge +2; ions 118 Na ⁺ , 120 Cl ⁻ ; water 18 845; net charge 0.
BMRB 10123; PDB 1N27	BMRB/PDB: PWWP domain [U- ¹³ C; U- ¹⁵ N], 0.7 mM; pH 6.0; 298 K; 90% H ₂ O/10% D ₂ O; 20 mM phosphate buffer, 100 mM NaCl, 1 mM d-DTT, 0.02% NaN ₃ ; $I = 120$ mM.	HIS24→HIE; HIS52→HIP; <i>Termini</i> : NGLY1/CGLY96; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.00$ 34 titratable; 1 non-default; 0 marginal $I = 122.32$ mM, from NaCl 100.00 and phosphate 22.32 phosphate as 22.32 mM NaCl $I_{\text{bulk}} = 123.21$ mM (genion : 51 NA, 54 CL) $I = 126.83$ mM after neutralisation	Solute charge +3; ions 51 Na ⁺ , 54 Cl ⁻ ; water 21 917; net charge 0.
BMRB 10138; PDB 2DOG	BMRB: RIEF fold [U- ¹³ C; U- ¹⁵ N], 1.2 mM; pH 6.5; 298 K; 90% H ₂ O/10% D ₂ O; 20 mM sodium phosphate; stated $I = 120$ mM.	HIS29→HIE; HIS38→HIE; HIS56→HIE; <i>Termini</i> : NMET1/CGLU85; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.50$ 31 titratable; 0 non-default; 1 marginal $I = 26.53$ mM, from phosphate 26.53 phosphate as 26.53 mM NaCl $I_{\text{bulk}} = 26.22$ mM (genion : 13 NA, 10 CL) $I = 30.16$ mM after neutralisation	Solute charge −3; ions 13 Na ⁺ , 10 Cl ⁻ ; water 20 297; net charge 0.
BMRB 10212; PDB 2CUD	BMRB/PDB: SH3 domain [U- ¹³ C; U- ¹⁵ N], 1.28 mM; pH 7.0; 298 K; 90% H ₂ O/10% D ₂ O; 20 mM d-TrisHCl, 100 mM NaCl, 1 mM d-DTT, 0.02% NaN ₃ ; stated $I = 120$ mM.	<i>Retained</i> : CYS69 free thiol; <i>Termini</i> : NGLY1/CGLY79; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 7.00$ 20 titratable; 0 non-default; 0 marginal $I = 118.40$ mM, from NaCl 100.00 and Tris 18.40 Tris as 18.40 mM NaCl $I_{\text{bulk}} = 117.79$ mM (genion : 74 NA, 72 CL) $I = 119.42$ mM after neutralisation	Solute charge −2; ions 74 Na ⁺ , 72 Cl ⁻ ; water 32 783; net charge 0.
BMRB 10240; PDB 1X69	BMRB/PDB: SH3 domain [U- ¹³ C; U- ¹⁵ N], 1 mM; pH 7.0; 296 K; 90% H ₂ O/10% D ₂ O; 20 mM d-Tris HCl, 100 mM NaCl, 1 mM d-DTT, 0.02% NaN ₃ ; stated $I = 120$ mM.	<i>Retained</i> : CYS57 free thiol; <i>Termini</i> : NGLY1/CGLY79; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 7.00$ 27 titratable; 0 non-default; 0 marginal $I = 118.40$ mM, from NaCl 100.00 and Tris 18.40 Tris as 18.40 mM NaCl $I_{\text{bulk}} = 118.49$ mM (genion : 95 NA, 84 CL) $I = 126.25$ mM after neutralisation	Solute charge −11; ions 95 Na ⁺ , 84 Cl ⁻ ; water 37 949; net charge 0.
BMRB 10241; PDB 1X6B	BMRB/PDB: SH3 domain [U- ¹³ C; U- ¹⁵ N], 1 mM; pH 7.0; 298 K; 90% H ₂ O/10% D ₂ O; 20 mM d-Tris HCl, 100 mM NaCl, 1 mM d-DTT, 0.02% NaN ₃ ; stated $I = 120$ mM.	<i>Termini</i> : NGLY1/CGLY79; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 7.00$ 20 titratable; 0 non-default; 0 marginal $I = 118.40$ mM, from NaCl 100.00 and Tris 18.40 Tris as 18.40 mM NaCl $I_{\text{bulk}} = 119.11$ mM (genion : 59 NA, 53 CL) $I = 125.85$ mM after neutralisation	Solute charge −6; ions 59 Na ⁺ , 53 Cl ⁻ ; water 23 586; net charge 0.

Record	Provided conditions	Protein preparation	R calculations		GROMACS
BMRB 10252; PDB 2CO9	BMRB/PDB: HMG-box domain [U- ¹³ C; U- ¹⁵ N], 1.28 mM; pH 7.0; 298 K; 90% H ₂ O/10% D ₂ O; 20 mM d-Tris-HCl, 100 mM NaCl, 1 mM d-DTT, 0.02% NaN ₃ ; stated $I = 120$ mM.	<i>Termini</i> : NGLY1/CGLY102; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i>	$pH_{\text{eff}} = 7.00$ 35 titratable; 0 non-default; 0 marginal $I = 118.40$ mM, from NaCl 100.00 and Tris 18.40 Tris as 18.40 mM NaCl $I_{\text{bulk}} = 118.05$ mM (genion : 84 NA, 93 CL) $I = 124.38$ mM after neutralisation	Solute charge +9; ions 84 Na ⁺ , 93 Cl ⁻ ; water 37 904; net charge 0.
BMRB 10261; PDB 2DJV	BMRB/PDB: WHEP-TRS domain [U- ¹³ C; U- ¹⁵ N], 1 mM; pH 7.0; 296 K; 90% H ₂ O/10% D ₂ O; 20 mM d-Tris HCl, 100 mM NaCl, 1 mM d-DTT, 0.02% NaN ₃ ; stated $I = 120$ mM.	<i>Termini</i> : NGLY1/CGLY79; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i>	$pH_{\text{eff}} = 7.00$ 25 titratable; 0 non-default; 0 marginal $I = 118.40$ mM, from NaCl 100.00 and Tris 18.40 Tris as 18.40 mM NaCl $I_{\text{bulk}} = 119.01$ mM (genion : 88 NA, 94 CL) $I = 123.06$ mM after neutralisation	Solute charge +6; ions 88 Na ⁺ , 94 Cl ⁻ ; water 39 680; net charge 0.
BMRB 10281; PDB 2CUE	BMRB/PDB: homeobox domain [U- ¹³ C; U- ¹⁵ N], 1.50 mM; pH 7.0; 298 K; 90% H ₂ O/10% D ₂ O; 20 mM d-TrisHCl, 100 mM NaCl, 1 mM d-DTT, 0.02% NaN ₃ ; stated $I = 120$ mM.	HIS30→HIE; <i>Termini</i> : NGLY1/CGLY80; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i>	$pH_{\text{eff}} = 7.00$ 31 titratable; 0 non-default; 0 marginal $I = 118.40$ mM, from NaCl 100.00 and Tris 18.40 Tris as 18.40 mM NaCl $I_{\text{bulk}} = 117.79$ mM (genion : 62 NA, 68 CL) $I = 123.49$ mM after neutralisation	Solute charge +6; ions 62 Na ⁺ , 68 Cl ⁻ ; water 28 259; net charge 0.
BMRB 10287; PDB 2DA5	BMRB/PDB: second homeobox domain [U- ¹³ C; U- ¹⁵ N], 1.2 mM; pH 7.0; 298 K; 90% H ₂ O/10% D ₂ O; 20 mM d-Tris-HCl, 100 mM NaCl, 1 mM d-DTT, 0.02% NaN ₃ ; stated $I = 120$ mM.	<i>Termini</i> : NGLY1/CGLY75; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i>	$pH_{\text{eff}} = 7.00$ 30 titratable; 0 non-default; 0 marginal $I = 118.40$ mM, from NaCl 100.00 and Tris 18.40 Tris as 18.40 mM NaCl $I_{\text{bulk}} = 119.09$ mM (genion : 50 NA, 52 CL) $I = 121.47$ mM after neutralisation	Solute charge +2; ions 50 Na ⁺ , 52 Cl ⁻ ; water 22 345; net charge 0.
BMRB 10289; PDB 2DA7	BMRB/PDB: homeobox domain [U- ¹³ C; U- ¹⁵ N], 1.2 mM; pH 7.0; 296 K; 90% H ₂ O/10% D ₂ O; 20 mM d-Tris-HCl, 100 mM NaCl, 1 mM d-DTT, 0.02% NaN ₃ ; stated $I = 120$ mM.	HIS16→HIE; <i>Termini</i> : NGLY1/CGLY71; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i>	$pH_{\text{eff}} = 7.00$ 21 titratable; 0 non-default; 0 marginal $I = 118.40$ mM, from NaCl 100.00 and Tris 18.40 Tris as 18.40 mM NaCl $I_{\text{bulk}} = 117.87$ mM (genion : 56 NA, 56 CL) $I = 117.87$ mM after neutralisation	Solute charge 0; ions 56 Na ⁺ , 56 Cl ⁻ ; water 25 062; net charge 0.
BMRB 10290; PDB 2DJN	BMRB/PDB: homeobox domain [U- ¹³ C; U- ¹⁵ N], 1.35 mM; pH 7.0; 298 K; 90% H ₂ O/10% D ₂ O; 20 mM d-Tris-HCl, 100 mM NaCl, 1 mM d-DTT, 0.02% NaN ₃ ; stated $I = 120$ mM.	<i>Termini</i> : NGLY1/CGLY70; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i>	$pH_{\text{eff}} = 7.00$ 18 titratable; 0 non-default; 0 marginal $I = 118.40$ mM, from NaCl 100.00 and Tris 18.40 Tris as 18.40 mM NaCl $I_{\text{bulk}} = 118.35$ mM (genion : 42 NA, 53 CL) $I = 133.85$ mM after neutralisation	Solute charge +11; ions 42 Na ⁺ , 53 Cl ⁻ ; water 18 692; net charge 0.
BMRB 10291; PDB 2DMN	BMRB: homeobox domain [U- ¹³ C; U- ¹⁵ N], 1.1 mM; pH 7.0; 298 K; 90% H ₂ O/10% D ₂ O; 20 mM d-Tris-HCl, 100 mM NaCl, 1 mM d-DTT, 0.02% NaN ₃ ; $I = 120$ mM.	HIS29→HIE; <i>Termini</i> : NGLY1/CGLY83; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i>	$pH_{\text{eff}} = 7.00$ 28 titratable; 0 non-default; 0 marginal $I = 118.40$ mM, from NaCl 100.00 and Tris 18.40 Tris as 18.40 mM NaCl $I_{\text{bulk}} = 118.93$ mM (genion : 86 NA, 94 CL) $I = 124.47$ mM after neutralisation	Solute charge +8; ions 86 Na ⁺ , 94 Cl ⁻ ; water 38 512; net charge 0.
BMRB 10292; PDB 2DMP	BMRB/PDB: homeobox domain [U- ¹³ C; U- ¹⁵ N], 1.2 mM; pH 7.0; 298 K; 90% H ₂ O/10% D ₂ O; 20 mM d-Tris-HCl, 100 mM NaCl, 1 mM d-DTT, 0.02% NaN ₃ ; $I = 120$ mM.	<i>Termini</i> : NGLY1/CGLY89; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i>	$pH_{\text{eff}} = 7.00$ 30 titratable; 0 non-default; 0 marginal $I = 118.40$ mM, from NaCl 100.00 and Tris 18.40 Tris as 18.40 mM NaCl $I_{\text{bulk}} = 118.61$ mM (genion : 104 NA, 106 CL) $I = 119.75$ mM after neutralisation	Solute charge +2; ions 104 Na ⁺ , 106 Cl ⁻ ; water 45 269; net charge 0.

Record	Provided conditions	Protein preparation	R calculations		GROMACS
BMRB 10293; PDB 2DMQ	BMRB: homeobox domain [U- ¹³ C; U- ¹⁵ N], 1.2 mM; pH 7.0; 298 K; 90% H ₂ O/10% D ₂ O; 20 mM d-Tris-HCl, 100 mM NaCl, 1 mM d-DTT, 0.02% NaN ₃ ; $I = 120$ mM.	HIS16→HIE; HIS17→HIE; HIS30→HIE; <i>Termini</i> : NGLY1/CGLY80; <i>OXT</i> : present.	<i>R</i>	$pH_{\text{eff}} = 7.00$	Solute charge +13; ions 103 Na ⁺ , 116 Cl ⁻ ; water 45 280; net charge 0.
			<i>HH</i>	24 titratable; 0 non-default; 0 marginal	
			<i>Target</i>	$I = 118.40$ mM, from NaCl 100.00 and Tris 18.40	
			<i>Proxy</i>	Tris as 18.40 mM NaCl	
			<i>Achieved</i>	$I_{\text{bulk}} = 118.16$ mM (genion : 103 NA, 116 CL)	
BMRB 10295; PDB 2DMT	BMRB/PDB: homeobox domain [U- ¹³ C; U- ¹⁵ N], 1.0 mM; pH 7.0; 298 K; 90% H ₂ O/10% D ₂ O; 20 mM d-Tris-HCl, 100 mM NaCl, 1 mM d-DTT, 0.02% NaN ₃ ; $I = 120$ mM.	<i>Termini</i> : NGLY1/CGLY80; <i>OXT</i> : present.	<i>Final I</i>	$I = 125.61$ mM after neutralisation	Solute charge +10; ions 78 Na ⁺ , 88 Cl ⁻ ; water 34 938; net charge 0.
			<i>R</i>	$pH_{\text{eff}} = 7.00$	
			<i>HH</i>	27 titratable; 0 non-default; 0 marginal	
			<i>Target</i>	$I = 118.40$ mM, from NaCl 100.00 and Tris 18.40	
			<i>Proxy</i>	Tris as 18.40 mM NaCl	
BMRB 10296; PDB 2DMU	BMRB/PDB: homeobox domain [U- ¹³ C; U- ¹⁵ N], 1.3 mM; pH 7.0; 296 K; 90% H ₂ O/10% D ₂ O; 20 mM d-Tris-HCl, 100 mM NaCl, 1 mM d-DTT, 0.02% NaN ₃ ; $I = 120$ mM.	HIS10→HIE; HIS45→HIE; <i>Termini</i> : NGLY1/CGLY70; <i>OXT</i> : present.	<i>Achieved</i>	$I_{\text{bulk}} = 119.11$ mM (genion : 78 NA, 88 CL)	Solute charge +5; ions 52 Na ⁺ , 57 Cl ⁻ ; water 23 474; net charge 0.
			<i>Final I</i>	$I = 126.74$ mM after neutralisation	
			<i>R</i>	$pH_{\text{eff}} = 7.00$	
			<i>HH</i>	29 titratable; 0 non-default; 0 marginal	
			<i>Target</i>	$I = 118.40$ mM, from NaCl 100.00 and Tris 18.40	
BMRB 10297; PDB 2E19	BMRB/PDB: homeobox domain [U- ¹³ C; U- ¹⁵ N], 1.0 mM; pH 7.0; 296 K; 90% H ₂ O/10% D ₂ O; 20 mM d-TrisHCl, 100 mM NaCl, 1 mM d-DTT, 0.02% NaN ₃ ; $I = 120$ mM.	<i>Termini</i> : NGLY1/CSER64; <i>OXT</i> : present.	<i>Proxy</i>	Tris as 18.40 mM NaCl	Solute charge +1; ions 54 Na ⁺ , 55 Cl ⁻ ; water 24 155; net charge 0.
			<i>Achieved</i>	$I_{\text{bulk}} = 117.58$ mM (genion : 52 NA, 57 CL)	
			<i>Final I</i>	$I = 123.23$ mM after neutralisation	
			<i>R</i>	$pH_{\text{eff}} = 7.00$	
			<i>HH</i>	14 titratable; 0 non-default; 0 marginal	
BMRB 10302; PDB 2DBM	BMRB: SH3 domain [U- ¹³ C; U- ¹⁵ N], 1 mM; pH 7.0; 296 K; 90% H ₂ O/10% D ₂ O; 20 mM d-Tris HCl, 100 mM NaCl, 1 mM d-DTT, 0.02% NaN ₃ ; $I = 120$ mM.	HIS48→HIE; HIS50→HIE; HIS67→HIE; <i>Retained</i> : CYS9 and CYS10 free thiols; <i>Termini</i> : NGLY1/CGLY73; <i>OXT</i> : present.	<i>Target</i>	$I = 118.40$ mM, from NaCl 100.00 and Tris 18.40	Solute charge −9; ions 59 Na ⁺ , 50 Cl ⁻ ; water 22 312; net charge 0.
			<i>Proxy</i>	Tris as 18.40 mM NaCl	
			<i>Achieved</i>	$I_{\text{bulk}} = 119.59$ mM (genion : 59 NA, 50 CL)	
			<i>Final I</i>	$I = 130.35$ mM after neutralisation	
			<i>R</i>	$pH_{\text{eff}} = 7.00$	
BMRB 10329; PDB 2D7N	BMRB/PDB: Filamin domain [U- ¹³ C; U- ¹⁵ N], 1.19 mM; pH 7.0; 296 K; 90% H ₂ O/10% D ₂ O; 20 mM d-Tris-HCl, 100 mM NaCl, 1 mM d-DTT, 0.02% NaN ₃ ; $I = 120$ mM.	HIS60→HIE; HIS70→HIE; HIS87→HIE; <i>Termini</i> : NGLY1/CGLY93; <i>OXT</i> : present.	<i>HH</i>	22 titratable; 0 non-default; 0 marginal	Solute charge +3; ions 67 Na ⁺ , 70 Cl ⁻ ; water 30 519; net charge 0.
			<i>Target</i>	$I = 118.40$ mM, from NaCl 100.00 and Tris 18.40	
			<i>Proxy</i>	Tris as 18.40 mM NaCl	
			<i>Achieved</i>	$I_{\text{bulk}} = 117.76$ mM (genion : 67 NA, 70 CL)	
			<i>Final I</i>	$I = 120.39$ mM after neutralisation	
BMRB 11173; PDB 2DAG	BMRB/PDB: UBA domain [U- ¹³ C; U- ¹⁵ N], 1.05 mM; pH 7.0; 296.0 K; 90% H ₂ O/10% D ₂ O; 20 mM d-Tris-HCl, 100 mM NaCl, 1 mM d-DTT, 0.02% NaN ₃ ; $I = 120$ mM.	HIS48→HIP; <i>Retained</i> : CYS26 free thiol; <i>Termini</i> : NGLY1/CGLY74; <i>OXT</i> : present.	<i>R</i>	$pH_{\text{eff}} = 7.00$	Solute charge −5; ions 64 Na ⁺ , 59 Cl ⁻ ; water 26 101; net charge 0.
			<i>HH</i>	15 titratable; 1 non-default; 1 marginal	
			<i>Target</i>	$I = 118.40$ mM, from NaCl 100.00 and Tris 18.40	
			<i>Proxy</i>	Tris as 18.40 mM NaCl	
			<i>Achieved</i>	$I_{\text{bulk}} = 117.63$ mM (genion : 64 NA, 59 CL)	
BMRB 11322; PDB 2CU7	BMRB/PDB: SANT domain [U- ¹³ C; U- ¹⁵ N], 0.83 mM; pH 4.5; 298 K; 90% H ₂ O/10% D ₂ O; 20 mM sodium acetate buffer, 100 mM NaCl, 1 mM d-DTT, 0.02% NaN ₃ ; $I = 120$ mM.	GLU15→GLH; GLU16→GLH; GLU18→GLH; GLU65→GLH; <i>Retained</i> : CYS60 free thiol; <i>Termini</i> : NGLY1/CGLY72; <i>OXT</i> : present.	<i>Final I</i>	$I = 122.62$ mM after neutralisation	Solute charge +13; ions 66 Na ⁺ , 79 Cl ⁻ ; water 32 870; net charge 0.
			<i>R</i>	$pH_{\text{eff}} = 4.50$	
			<i>HH</i>	26 titratable; 4 non-default; 3 marginal	
			<i>Target</i>	$I = 107.09$ mM, from NaCl 100.00 and acetate 7.09	
			<i>Proxy</i>	sodium acetate as 7.09 mM NaCl	
			<i>Achieved</i>	$I_{\text{bulk}} = 107.28$ mM (genion : 66 NA, 79 CL)	
			<i>Final I</i>	$I = 117.85$ mM after neutralisation	

Record	Provided conditions	Protein preparation	R calculations		GROMACS
BMRB 15129; PDB 2NWM	BMRB: Vinexin SH3 domain [U- ¹³ C; U- ¹⁵ N], 1–1.2 mM; pH 6.2; 298 K; 90% H ₂ O/10% D ₂ O; 50 mM phosphate buffer, 50 mM NaCl, 1 mM DTT, 1 mM EDTA.	HIS29→HIP; HIS41→HIP; HIS42→HIE; HIS60→HIP; HIS61→HIP; HIS62→HIP; HIS63→HIP; HIS64→HIP; <i>Termini</i> : NMET1/CHIP65; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.20$ 29 titratable; 8 non-default; 8 marginal $I = 113.52$ mM, from phosphate 58.90, NaCl 50.00, and EDTA 4.62 phosphate and EDTA as 63.52 mM NaCl $I_{\text{bulk}} = 112.94$ mM (genion : 47 NA, 54 CL) $I = 121.35$ mM after neutralisation	Solute charge +7; ions 47 Na ⁺ , 54 Cl [−] ; water 22 197; net charge 0.
BMRB 15133; PDB 2JNY	BMRB: Uncharacterized BCR [U- ¹³ C; U- ¹⁵ N], 1.0 mM; pH 6.5; 298 K; 95% H ₂ O/5% D ₂ O; 20 mM MES, 100 mM NaCl, 5 mM CaCl ₂ , 100 mM DTT, 0.02% NaN ₃ ; stated $I = 100$ mM.	<i>Retained</i> : CYS13 free thiol; <i>Termini</i> : NMET1/CGLU61; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.50$ 21 titratable; 0 non-default; 0 marginal $I = 120.69$ mM, from NaCl 100.00, MES 5.69, and CaCl ₂ 15.00 MES as 5.69 mM NaCl $I_{\text{bulk}} = 123.09$ mM (genion : 2 CA; 42 NA, 39 CL) $I = 131.99$ mM after neutralisation	Solute charge −7; ions 2 Ca ²⁺ , 42 Na ⁺ , 39 Cl [−] ; water 17 879; net charge 0.
BMRB 15142; PDB 2E8J	BMRB: Dynein light chain 2A [U- ¹³ C; U- ¹⁵ N], 0.6 mM; pH 7.0; 298 K; 95% H ₂ O/5% D ₂ O; 20 mM phosphate buffer, 20 mM NaCl, 0.1 mM EDTA; stated $I = 40$ mM.	HIS52→HIE; <i>Termini</i> : NGLY1/CGLU101; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 7.00$ 28 titratable; 0 non-default; 0 marginal $I = 55.82$ mM, from phosphate 35.26, NaCl 20.00, and EDTA 0.56 phosphate and EDTA as 35.82 mM NaCl $I_{\text{bulk}} = 55.69$ mM (genion : 26 NA, 26 CL) $I = 55.69$ mM after neutralisation	Solute charge 0; ions 26 Na ⁺ , 26 Cl [−] ; water 24 567; net charge 0.
BMRB 15167; PDB 2JOE	BMRB: YehR [U- ¹³ C; U- ¹⁵ N], 1.0 mM; pH 5.5; 283 K; 95% H ₂ O/5% D ₂ O; 20 mM NH ₄ OAc, 100 mM NaCl, 5 mM CaCl ₂ , 10 mM DTT, 0.02% NaN ₃ ; stated $I = 100$ mM.	GLU82→GLH; HIS134→HIP; HIS135→HIP; HIS136→HIP; HIS137→HIP; HIS138→HIP; <i>Termini</i> : NMET1/CHIP139; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 5.50$ 52 titratable; 7 non-default; 0 marginal $I = 135.00$ mM, from CaCl ₂ 15.00, NaCl 100.00, and ammonium acetate 20.00 ammonium acetate as 20.00 mM NaCl $I_{\text{bulk}} = 134.73$ mM (genion : 3 CA; 70 NA, 81 CL) $I = 138.99$ mM after neutralisation	Solute charge +5; ions 3 Ca ²⁺ , 70 Na ⁺ , 81 Cl [−] ; water 30 976; net charge 0.
BMRB 15324; PDB 2JRA	BMRB/PDB: RPA2121 [U-99% ¹³ C; U-99% ¹⁵ N], 0.5 mM; pH 7.0; 298 K; 90% H ₂ O/10% D ₂ O; 10 mM d-Tris, 300 mM NaCl, 0.01% NaN ₃ , 1 mM benzamidine; $I = 300$ mM.	HIS48→HIE; <i>Termini</i> : NMET1/CLYS67; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 7.00$ 20 titratable; 0 non-default; 0 marginal $I = 309.20$ mM, from NaCl 300.00 and Tris 9.20 Tris as 9.20 mM NaCl $I_{\text{bulk}} = 308.78$ mM (genion : 299 NA, 300 CL) $I = 309.30$ mM after neutralisation	Solute charge +1; ions 299 Na ⁺ , 300 Cl [−] ; water 52 227; net charge 0.
BMRB 15332; PDB 2JRH	BMRB/PDB: MEKK3 PB1-cis [U- ¹³ C; U- ¹⁵ N], 0.5 mM; pH 6.0; 298 K; 90% H ₂ O/10% D ₂ O; 50 mM phosphate buffer, 1 mM EDTA; stated $I = 50$ mM.	HIS11→HIP; HIS31→HIP; HIS44→HIE; HIS89→HIP; HIS90→HIP; HIS91→HIP; HIS92→HIP; HIS93→HIP; <i>Termini</i> : NMET1/CHIP94; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.00$ 40 titratable; 8 non-default; 1 marginal $I = 60.08$ mM, from phosphate 55.80 and EDTA 4.28 phosphate and EDTA as 60.08 mM NaCl $I_{\text{bulk}} = 60.42$ mM (genion : 33 NA, 39 CL) $I = 65.91$ mM after neutralisation	Solute charge +6; ions 33 Na ⁺ , 39 Cl [−] ; water 29 187; net charge 0.
BMRB 15339; PDB 2JRM	BMRB: VpR55 [U- ¹³ C; U- ¹⁵ N], 1.1 mM; pH 6.5; 95% H ₂ O/5% D ₂ O; 20 mM MES, 100 mM NaCl, 5 mM CaCl ₂ , 10 mM DTT, 0.02% NaN ₃ ; stated $I = 100$ mM. PDB: 293 K.	HIS60→HIE; HIS61→HIP; HIS62→HIE; HIS63→HIE; HIS64→HIE; <i>Retained</i> : CYS29 free thiol; <i>Termini</i> : NMET1/CHIE65; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.50$ 33 titratable; 1 non-default; 4 marginal $I = 120.69$ mM, from MES 5.69, NaCl 100.00, and CaCl ₂ 15.00 MES as 5.69 mM NaCl $I_{\text{bulk}} = 121.24$ mM (genion : 2 CA; 41 NA, 51 CL) $I = 130.37$ mM after neutralisation	Solute charge +6; ions 2 Ca ²⁺ , 41 Na ⁺ , 51 Cl [−] ; water 20 410; net charge 0.
BMRB 15341; PDB 2JRR	BMRB: SiR90 [5% ¹³ C; U- ¹⁵ N] sample; pH 6.5; 293 K; 95% H ₂ O/5% D ₂ O; 10 mM Tris, 100 mM NaCl, 5 mM DTT; stated $I = 100$ mM.	CYS18→CYM; HIS27→HIP; HIS62→HIP; HIS63→HIE; HIS64→HIP; HIS65→HIE; HIS66→HIP; <i>Retained</i> : CYS43, CYS46, and CYS48 free thiols; <i>Termini</i> : NMET1/CHIP67; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.50$ 31 titratable; 6 non-default; 6 marginal $I = 109.73$ mM, from Tris 9.73 and NaCl 100.00 Tris as 9.73 mM NaCl $I_{\text{bulk}} = 109.04$ mM (genion : 37 NA, 38 CL) $I = 110.51$ mM after neutralisation	Solute charge +1; ions 37 Na ⁺ , 38 Cl [−] ; water 17 992; net charge 0.

Record	Provided conditions	Protein preparation	R calculations		GROMACS
BMRB 15355; PDB 2PPH	BMRB/PDB: MEKK3 PB1 [U- ¹³ C; U- ¹⁵ N], 0.5 mM; pH 6.0; 298 K; 90% H ₂ O/10% D ₂ O; 50 mM phosphate buffer, 1 mM EDTA.	HIS11→HIP; HIS31→HIP; HIS44→HIE; HIS89→HIP; HIS90→HIP; HIS91→HIP; HIS92→HIP; HIS93→HIP; <i>Termini</i> : NMET1/CHIP94; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.00$ 40 titratable; 8 non-default; 1 marginal $I = 60.08$ mM, from phosphate 55.80 and EDTA 4.28 phosphate and EDTA as 60.08 mM NaCl $I_{\text{bulk}} = 60.54$ mM (genion : 33 NA, 39 CL) $I = 66.04$ mM after neutralisation	Solute charge +6; ions 33 Na ⁺ , 39 Cl ⁻ ; water 29 134; net charge 0.
BMRB 15410; PDB 2LD9	BMRB: human ubiquitin [U- ¹³ C; U- ¹⁵ N], 1.7 mM; pH 4.6; 298 K; 90% H ₂ O/10% D ₂ O; 20 mM phosphate; $I = 20$ mM.	GLU18→GLH; GLU24→GLH; GLU51→GLH; HIS68→HIP; <i>Termini</i> : NMET1/CGLY76; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 4.60$ 25 titratable; 4 non-default; 4 marginal $I = 20.03$ mM, from phosphate 20.03 phosphate as 20.03 mM NaCl $I_{\text{bulk}} = 19.70$ mM (genion : 7 NA, 11 CL) $I = 25.33$ mM after neutralisation	Solute charge +4; ions 7 Na ⁺ , 11 Cl ⁻ ; water 18 779; net charge 0.
BMRB 15531; PDB 1CSG	BMRB: rhGM-CSF [U- ¹³ C; U- ¹⁵ N], 0.4 mM; pH 7.4; 298 K; 95% H ₂ O/5% D ₂ O; 2 mM sodium phosphate, 1 mM DSS; $I = 2$ mM.	HIS15→HIE; HIS83→HIE; HIS87→HIE; <i>Bonded</i> : CYS54–CYS96 and CYS88–CYS121 S–S; residues CYX; <i>Termini</i> : NALA1/CGLU127; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 7.40$ 39 titratable; 4 non-default; 0 marginal $I = 4.43$ mM, from phosphate 4.43 phosphate as 4.43 mM NaCl $I_{\text{bulk}} = 3.77$ mM (genion : 7 NA, 2 CL) $I = 8.48$ mM after neutralisation	Solute charge −5; ions 7 Na ⁺ , 2 Cl ⁻ ; water 28 622; net charge 0.
BMRB 15591; PDB 1Q02	BMRB: P62 UBA domain [U- ¹³ C; U- ¹⁵ N], 1 mM; pH 7; 298 K; 90% H ₂ O/10% D ₂ O; 50 mM potassium phosphate, 50 mM NaCl, 0.04% NaN ₃ .	<i>Termini</i> : NGLY1/CHIE52; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 7.00$ 15 titratable; 0 non-default; 0 marginal $I = 138.14$ mM, from NaCl 50.00 and phosphate 88.14 phosphate as 88.14 mM NaCl $I_{\text{bulk}} = 139.48$ mM (genion : 44 NA, 41 CL) $I = 144.58$ mM after neutralisation	Solute charge −3; ions 44 Na ⁺ , 41 Cl ⁻ ; water 15 542; net charge 0.
BMRB 15756; PDB 2CG7	BMRB: 2F13F1 [U- ¹³ C; U- ¹⁵ N], 0.3 mM; pH 6.0; 303 K; 95% H ₂ O/5% D ₂ O; 20 mM sodium phosphate, 20 mM NaCl, 0.01% NaN ₃ , 0.1 mM DSS; $I = 115$ mM.	HIS49→HIE; HIS65→HIP; <i>Bonded</i> : CYS4–CYS32, CYS30–CYS42, CYS48–CYS76, and CYS74–CYS86 S–S; residues CYX; <i>Termini</i> : NGLU1/CILE89; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.00$ 40 titratable; 9 non-default; 1 marginal $I = 115.00$ mM, from a deposited scalar value phosphate plus scalar gap as 95.00 mM NaCl $I_{\text{bulk}} = 115.11$ mM (genion : 67 NA, 68 CL) $I = 115.97$ mM after neutralisation	Solute charge +1; ions 67 Na ⁺ , 68 Cl ⁻ ; water 31 188; net charge 0.
BMRB 15784; PDB 2BA3	BMRB/PDB: NikA(1–51) [U- ¹³ C; U- ¹⁵ N], 2 mM; pH 6.8; 298 K; 90% H ₂ O/10% D ₂ O; 50 mM phosphate buffer, 50 mM NaCl, 0.02% w/v NaN ₃ ; $I = 50$ mM.	<i>Termini</i> : NSER1/CASN51; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.80$ 20 titratable; 0 non-default; 0 marginal $I = 128.01$ mM, from phosphate 78.01 and NaCl 50.00 phosphate as 78.01 mM NaCl $I_{\text{bulk}} = 128.08$ mM (genion : 92 NA, 96 CL) $I = 130.86$ mM after neutralisation	Solute charge +4; ions 92 Na ⁺ , 96 Cl ⁻ ; water 38 427; net charge 0.
BMRB 15884; PDB 1DIV	BMRB: CTL9 [U- ¹³ C; U- ¹⁵ N], 1 mM; pH 3.8; 298.2 K; 90% H ₂ O/10% D ₂ O; 20 mM sodium acetate, 100 mM NaCl; $I = 120 \pm 10$ mM.	GLU3→GLH; GLU4→GLH; GLU13→GLH; GLU28→GLH; GLU43→GLH; HIS49→HIP; GLU59→GLH; ASP62→ASH; HIS77→HIP; GLU79→GLH; HIS87→HIP; GLU90→GLH; <i>Termini</i> : NALA1/CLYS92; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 3.80$ 31 titratable; 12 non-default; 2 marginal $I = 101.98$ mM, from NaCl 100.00 and acetate 1.98 sodium acetate as 1.98 mM NaCl $I_{\text{bulk}} = 102.10$ mM (genion : 55 NA, 71 CL) $I = 116.95$ mM after neutralisation	Solute charge +16; ions 55 Na ⁺ , 71 Cl ⁻ ; water 28 919; net charge 0.
BMRB 16989; PDB 2CQB	BMRB: CYP33 RRM domain; pH 6.9; 303 K; 95% H ₂ O/5% D ₂ O; 25 mM potassium phosphate, 50 mM NaCl, 1 mM DTT, 1 mM NaN ₃ .	HIS23→HIE; HIS46→HIE; <i>Termini</i> : NGLY1/CLYS83; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 6.90$ 29 titratable; 0 non-default; 0 marginal $I = 91.44$ mM, from NaCl 50.00 and phosphate 41.44 phosphate as 41.44 mM NaCl $I_{\text{bulk}} = 91.60$ mM (genion : 40 NA, 32 CL) $I = 103.05$ mM after neutralisation	Solute charge −8; ions 40 Na ⁺ , 32 Cl ⁻ ; water 18 485; net charge 0.
BMRB 17439; PDB 1UBQ	BMRB: ubiquitin [U- ¹³ C; U- ¹⁵ N], 1.64 mM; pH 7.2; 298 K; 92.5% H ₂ O/7.5% D ₂ O; 75 mM sodium phosphate, 30 μ M DSS; $I = 185$ mM.	HIS68→HIE; <i>Termini</i> : NMET1/CGLY76; <i>OXT</i> : present.	<i>R</i> <i>HH</i> <i>Target</i> <i>Proxy</i> <i>Achieved</i> <i>Final I</i>	$pH_{\text{eff}} = 7.20$ 25 titratable; 0 non-default; 0 marginal $I = 149.14$ mM, from phosphate 149.14 phosphate as 149.14 mM NaCl $I_{\text{bulk}} = 149.18$ mM (genion : 53 NA, 53 CL) $I = 149.18$ mM after neutralisation	Solute charge 0; ions 53 Na ⁺ , 53 Cl ⁻ ; water 18 685; net charge 0.